Parse-x: A Literature Review for the Deep Benchmark Framework

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Abstract

Document-parsing evaluation has bifurcated into two literatures with little overlap: a multimodal-LLM benchmark line (OCRBench v2, OmniDocBench, olmOCR-Bench) that reports single aggregate numbers per model, and an in-house or release-discipline benchmark line (SAVIOR-Bench; the Apache-2.0 ParseBench corpus from LlamaIndex re-used with project-specific evaluation discipline) that exposes per-document score vectors, multilingual-quality stratification, and downstream-task lift. This review surveys the prior art for each of the seven measurement dimensions defined by the Parse-x Deep Benchmark Framework (D-OCR, D-PARSE-INTENT, D-LAYOUT, D-BBOX, D-TABLE, D-DOWNSTREAM, D-COST-LATENCY), together with the two stratification overlays (D-DOC-QUALITY and D-LANG) that cross-cut them and the sixteen-document-type taxonomy that cross-cuts them. For each dimension, the canonical benchmarks, relevant model families, and specific deltas that motivate the framework's additional measurement discipline are identified. Three appendices provide a structured model-lineage table covering the 50 or more models in the project's measurement pool, a datasets-and-contamination posture register, and a vendor-and-API pin table for the closed-source services in scope. The review adopts a descriptive rather than evaluative stance: prior work is named for what it measures and what it does not measure, no winner is selected in advance, and most quantitative claims are anchored to numbered references, with the few that remain inline-flagged as unverified. See also the recent table-recognition survey [167] and the 2024-2025 Vision-Language Model (VLM)-era document-understanding survey lineage [168, 169, 171] for adjacent taxonomies that this review extends with per-dimension scoring contracts and stratification overlays.

Contents

1. Introduction

The Parse-x Deep Benchmark Framework [25] decomposes the question of whether a given document-parsing engine is suitable for finance documents into nine orthogonal measurement dimensions and a sixteen-type document taxonomy. Each cell of the resulting matrix is anchored, where possible, to a public benchmark (ParseBench [49], SAVIOR-Bench [1], OmniDocBench [4], PubTabNet, DocLayNet) and falls back to in-house silver pipelines (BBOX-SILVER-PIPELINE-SPEC) or stratified hold-outs (dataset_95) when public coverage is absent. The framework's publish-gate requires per-document score vectors with bootstrap CI95 (or an explicit attestation label), uniform decoding pins (temperature 0.1, top-p 0.9, max_tokens 2048, seed 20260516), and fine-tune-vs-zero-shot disclosure on every cell.

This review serves two purposes. First, it surveys the prior art per dimension so that each new leaderboard cell can be interpreted against an appropriate historical baseline. Second, it identifies methodological gaps in that prior art (notably the absence of per-document vectors, inconsistent fine-tune disclosure, under-stratified multilingual and quality axes, and the near-total absence of parse-to-downstream lift measurements) that motivate the framework's additional measurement discipline. Per [161] and [162], per-document score vectors with bootstrap confidence intervals are now a baseline expectation for language-model evaluation; the contribution claimed here is not methodological novelty over that baseline but rather an industrial framework and curated dataset registry that ports the discipline to document parsing, exposes the open gates, and provides a rerunnable-cell convention compatible with [163] (DUE, the most relevant prior document-understanding framework). Earlier surveys [159, 160] catalogue the field at the model level; this review pairs that prior art with a measurement-level framework.

1.1 Contributions

The contributions of this review are operational rather than methodological, and may be summarized as follows:

1. **A nine-axis coverage map** of the document-parsing literature 2019–2026, comprising seven measurement dimensions and two stratification overlays, with per-dimension prior-art catalogues.

2. **A unified model-lineage table** (Appendix A) covering more than 50 open-source and proprietary engines, with base architecture, training data, license, and version pin per row.

3. **A datasets-and-contamination posture register** (Appendix B) identifying which corpora are public or internal and flagging known training-data overlap.

4. **A vendor-and-API pin table** (Appendix C) capturing the closed-source services in scope, with current model snapshots and API version pins as of 2026-06-16.

5. **A residuals list** (§4) enumerating reproducibility blockers, statistical-power limits, and four content surfaces (handwriting, mathematical formula extraction, document classification, and long-document handling) treated only briefly.

6. **A measurement-discipline template** ported from HELM [161] and BIG-bench [162] and applied to document parsing as a standing release-gate rather than a per-paper choice.

Running in parallel to that academic line is a substantial industrial-IDP literature (UiPath Document Understanding [172], Hyperscience [175], Rossum [176], Microsoft 365 document processing [177], Google Document AI [174], AWS Textract [173], plus the Forrester/Gartner/Everest/IDC analyst taxonomies that segment them) which shares the parse-then-extract architecture but reports customer-pilot headline accuracy rather than per-document vectors; §2.8 surveys that landscape and its evaluation-discipline gap.

The remainder of the paper follows the structure of the framework. §2 surveys the seven dimensions in order, with the D-DOC-QUALITY and D-LANG stratifications applied as overlays throughout. Appendices A–C provide the model lineage table, the datasets-and-contamination posture register, and the vendor-and-API pin table. Section 3 closes with a discussion of the principal open gates that block first-pass cell publication.

2. The Seven Measurement Dimensions and Two Stratification Overlays

D-OCR: text recognition transcription

The D-OCR dimension of Parse-x scores the most primitive capability of any document-AI engine: given a page image or PDF, the recovery of the document's tokens and the fidelity of that recovery. The framework's primary metric is `savior_word_recall`, a multiset recall over reference tokens taken from SAVIOR-Bench (n=509, Inter-Annotator Agreement (IAA) 0.761 via fuzzy-string similarity over 187 double-annotated pairs), with `savior_word_precision`, normalized Levenshtein `ocr_edit_distance`, and word error rate as secondaries [1][2]. The dimension is separated from D-PARSE-INTENT (structured field extraction) and D-DOWNSTREAM (chained pipelines) so that hallucinations and dropped tokens surface as transcription-level signals rather than being masked by a downstream rubric. The publish-gate requires per-document score vectors (for 2000-resample bootstrap CI95) or an explicit `provenance: attested` label; a discipline that most external OCR leaderboards do not currently satisfy.

Prior benchmarks

OCRBench v2 [3] is the most widely cited LMM-era OCR benchmark: 10,000 human-verified QA pairs across 23 tasks and 31 scenarios (street scenes, receipts, formulas, diagrams), with a 1,500-image private split. Its headline finding, that 20 of 22 evaluated LMMs score below 50/100, indicates that OCR remains far from saturated for multimodal models. However, OCRBench v2 is a QA-style benchmark (scored with ANLS-family metrics in the DocVQA lineage [209]): it measures whether a model can *answer questions about text*, not whether it produces a full faithful transcription. Word-level recall, edit distance, and verbose-hallucination WER patterns (the failure mode SAVIOR-Bench surfaces for PaddleOCR-VL-1.5 and DeepSeek-OCR-2) are not its primary signal [1].

OmniDocBench [4], published at CVPR 2025, supplies among the most comprehensive *document-parsing* benchmarks in the prior art: nine document sources (academic papers, textbooks, handwritten notes, dense newspapers), 19 layout categories, 15 attribute labels, and end-to-end plus task-specific evaluation modes.

It is the canonical leaderboard cited by MinerU2.5 (overall 90.67) [5] and PaddleOCR-VL-1.5 (94.5% on v1.5) [6]. OmniDocBench measures text + layout + table jointly through end-to-end edit-distance-style metrics; it does not separate transcription quality from layout fidelity the way Parse-x splits D-OCR from D-LAYOUT, consequently, a system that emits clean text in incorrect reading order is scored together with one that drops tokens.

olmOCR-Bench [7] from AI2 covers 7,000+ test cases across 1,400 documents, with the design constraint that systems must emit plain-text Unicode in natural reading order. olmOCR-2-7B-1025 scores 82.4 on the benchmark [8]. olmOCR-Bench is English-print-centric and unit-test-rule structured (presence/absence/order checks), which is complementary to, but narrower than, SAVIOR-Bench's recall/precision/edit-distance triad over a multilingual low-quality-scan corpus.

SAVIOR-Bench v1 [1] (see App. B.1) is the in-house gold corpus on which the framework primarily anchors D-OCR; it is relevant here because its scoring discipline (per-doc vectors, IAA reporting) is stricter than that of public OCR benchmarks. However, the corpus is not publicly redistributed, and the per-doc vectors for the attested 31-cell leaderboard remain an open gate (R2 / BLK-18).

FUNSD, CORD, and the classical CER/WER leaderboards [9] (see App. B.1) remain useful for printed-form recognition but cover none of the modern failure surfaces (vertical text, logo-embedded text, multilingual + dense layout) that SAVIOR-Bench's failure taxonomy enumerates [1].

Models surveyed

The dossier inventories approximately 31 OCR cells in `DC-000__ocr-edit-distance-v1` [10], grouped by family as follows.

- **2025-2026 frontier additions.** xAI Grok 4 / 4.3 (multimodal with native OCR, 256k context; Grok 4.3 beta April 2026 adds native multimodal video) [190]; Meta Llama 4 family, Scout (109B / 10M-token context) and Maverick (400B / 1M-token context), both with native multimodality via early-fusion [191]. Both join the panel as candidate D-OCR engines once OCR-specific evaluation lands; current panel rows are PENDING-MEASUREMENT.
- **General-purpose document VLMs.** Qwen3-VL-235B, Qwen2.5-VL-3B/7B (the SAVIOR-Train fine-tune base), Qwen3.6-35B-A3B, InternVL2-8B/26B/76B, LLaVA-1.6-34B, GPT-4o / GPT-4o-mini, Gemini-1.5 Flash/Pro, Claude-Sonnet-4.6 [10]. Most are evaluated zero-shot; SAVIOR fine-tuning on Qwen2.5-VL-7B with LoRA (rank 32, α 64, 5 epochs, 2× H100) lifts word-recall to 0.9294 [2][1].
- **OCR-specialist VLMs.** GLM-OCR (edit-distance 0.019, rank-1 specialist on the SAVIOR leaderboard; sibling to the GLM-4.5 foundation series [205]), DeepSeek-OCR-2, GOT-OCR-2.0, DocOwl2 / mPLUG-DocOwl1.5, Fox, TextMonkey, Vary-toy, Nougat-0.1.0, FireRed-OCR, Logics-Parsing-v2, Chandra (`datalab-to/chandra-ocr-2`) [10][1].
- **Decoupled / pipeline parsers.** MinerU 0.9 and MinerU 2.5 (1.2B, coarse-to-fine decoupled, OmniDocBench 90.67) [5]; Marker-1.0; Surya; PaddleOCR-v4 and PaddleOCR-VL-1.5 (0.9B ERNIE-4.5 backbone, OmniDocBench v1.5 94.5%) [6]; HyperAPI-PaddleOCR-OURS; olmOCR / olmOCR-2-7B-1025 [8].
- **Classical engines.** Tesseract-OCR (multilingual restored 2026-06-10 via `tesseract-lang`), EasyOCR, OpenOCR; Mathpix as a hosted reference [10].
- **Cloud / vendor systems.** Textract (SAVIOR word-recall 0.9233, #1 zero-shot anchor) [1].

Gaps relative to the framework

First, prior benchmarks largely report *single aggregate numbers per model* and do not retain per-document score vectors; OCRBench v2, OmniDocBench, and the SAVIOR attested leaderboard all currently lack the per-doc vectors Parse-x requires for bootstrap CI95 [3][4][1]. Second, the verbose-hallucination pattern (PaddleOCR-VL-1.5: F1 0.726 with WER 2.96; DeepSeek-OCR-2: WER 12.35) only emerges when WER is reported alongside F1 across the same cell, a discipline OCRBench v2's QA framing structurally cannot expose [1]. Third, no prior benchmark stratifies D-OCR by the Q-LOW/Q-MED/Q-HIGH quality axis the framework defines under D-DOC-QUALITY, so a model's collapse on degraded scans is reported only anecdotally in failure-mode lists [1][2]. Fourth, the multilingual axis is underspecified everywhere except OmniDocBench's nine-source mix and PaddleOCR-VL-1.5's Tibetan/Bengali extension [4][6]; Parse-x's D-LANG cross-stratification of D-OCR appears, on the current literature, to be novel. Finally, fine-tune-vs-zero-shot disclosure is inconsistent in public leaderboards (a recurring SAVIOR-Bench complaint, R6); the framework's `(fine-tuned)` tagging functions as both a measurement-disclosure and metric contribution [1].

D-PARSE-INTENT: structured field extraction F1 + WER

The D-PARSE-INTENT dimension scores a document engine's capacity to emit a canonical field-set with values from a raw page, without a hand-written schema. It is distinct from D-OCR (no transcription objective) and from D-DOWNSTREAM (no chained extractor LLM). The framework pins `parse_intent_f1` (per-field F1 over the recovered intent fields, ↑) as the primary metric and `wer` (word error rate on the intent-bearing text subset, ↓) as a secondary metric, exposing a recurring failure pattern in which verbose VLMs hallucinate around the value sufficiently to keep the field recoverable while transcription degrades (for example, PaddleOCR-VL-1.5 reaches F1 0.726 with WER 2.96 on the inhouse-v1 panel) [11]. Compared with D-OCR, the public prior art for D-PARSE-INTENT is markedly thinner: most benchmarks are either narrow-domain (receipts, forms) or vendor-curated, and few jointly publish a structured-F1 alongside a transcription-error metric on the same documents [1].

Prior benchmarks

- **FUNSD** [9] (see App. B.1): relevant here for form key-value labelling (entity-level SER F1) but does not exercise line-item structure, multi-page reasoning, or transcription error.
- **CORD** [13]: 1,000 Indonesian receipts (800/100/100), 30 entity types in 4 groups (Menu / Void menu / Subtotal / Total). Evaluated with field-level F1 and tree-edit-distance accuracy; LayoutLMv3-large reports 97.46 F1 (Table 1, official 800/100 split; base 96.56), Donut 91.6 [12][14]. The corpus is receipt-only, comprises short documents, and is monolingual.
- **SROIE (ICDAR 2019, Task 3)** [15] (see App. B.1): the de facto entry-level Key Information Extraction (KIE) benchmark; relevant here as a fixed-schema baseline well below the free-schema inhouse-v1 panel.
- **Kleister NDA / Kleister Charity** [16]: 540 NDAs (3,229 pages, 2,160 entities) and 2,788 charity reports (61,643 pages, 21,612 entities); strongest models report 81.77 / 83.57 F1; adds long-document layout, but only English and non-financial-invoice schemas.
- **DocILE-2023** [17]: 6.7k annotated business documents + 100k synthetic + ~1M unlabelled with 55 KILE/LIR classes from invoices/orders (ICDAR-2023 + CLEF-2023 competition); the closest public analogue to the framework's invoice-heavy T-01 doc-type, though it does not score transcription WER alongside KIE F1.

- **DeepForm** [18]: political-ad TV-station disclosure forms maintained by ProPublica; appears as a KIE task in the DUE benchmark [19] where answer normalization is required. The domain is narrow and the held-out partition is small.
- **DUE (Document Understanding Evaluation)** [19]: meta-benchmark assembling DocVQA, InfographicsVQA, Kleister, DeepForm, WikiTableQuestions and others under one QA-style harness; useful as a contamination reference because most modern VLMs have seen at least part of it.

The framework's own `parse-intent-inhouse-v1` rubric (23 attested systems; n unreported in the attested source [1]) is internal and intentionally not externalised. The dossier flags that the rubric provenance must not be derived from `results.zip` (HyperAPI deployed output), and a contamination check (Q-DOSS-2) remains open [11].

Models surveyed

- **Layout-aware encoders**: LayoutLMv3 (base + large) [12], BROS [20], LiLT [21], FormNetV2 (86.35 F1 on FUNSD at 2.6× smaller than DocFormer) [22].
- **OCR-free generative VDU**: Donut (CORD 91.6) [14] and the OmniParser line (text-spotting + KIE + table) [23]. An important precursor is **TILT** (Powalski et al., ICDAR 2021) [157], a text-image-layout encoder-decoder that reported state-of-the-art on DocVQA, CORD and SROIE and established the joint multimodal-transformer recipe later inherited by Donut and UDOP.
- **General multimodal foundation models** evaluated on the inhouse-v1 panel: GPT-4o (well-documented zero-shot anchor at F1 0.8052 / WER 0.4232), GPT-4.1, Claude-Sonnet-4.6, Gemini-2.5-Flash, Gemini-2.5-Pro, Llama-3.2-90B-Vision, Mistral-Large-3-675B, plus the 2025-2026 Mistral specialists Magistral (reasoning) [203] and Devstral (agentic SWE) [204] [11][1]. Per project guardrail [Hyperbots Azure OpenAI routing policy], OpenAI calls route through Azure OpenAI Service.
- **VLM/OCR specialists** also scored on inhouse-v1: Qwen2.5-VL-3B/7B-Instruct, Qwen3-VL-32B-Instruct, Qwen3.6-35B-A3B, PaddleOCR-VL-1.5, MinerU2.5-Pro, GLM-OCR, FireRed-OCR, Logics-Parsing-v2, DeepSeek-OCR-2, Qianfan-OCR, DocIntentOCR-3B (under-documented; Q-DF-3 open) [11].
- **Fine-tuned in-house**: `hyperapi-parse-intent (fine-tuned)` leads at F1 0.8328 / WER 0.2589 (+0.0276 F1 over the GPT-4o zero-shot anchor) [1].

Gaps relative to the framework

1. **Joint F1 + WER on identical inputs.** FUNSD/CORD/SROIE/Kleister/DocILE all report a single structured-F1; none publishes a paired transcription WER over the same intent-bearing token subset. The framework's secondary `wer` metric surfaces the "high-F1, inflated-WER" pattern (PaddleOCR-VL-1.5: F1 0.726, WER 2.96) [11].

2. **Schema-free task definition.** SROIE (4 fields) and CORD (30 entities) fix the schema in advance; inhouse-v1 measures whether the engine itself can propose the canonical field-set without a schema prompt, a capability prior public benchmarks do not score [1].

3. **Cross-vendor panel discipline.** Most public KIE papers report 2–6 baselines on one corpus; the inhouse-v1 panel runs 23 systems under pinned decoding with attestation [1], matching the apples-to-apples gate but inheriting the contamination question on the rubric itself [11].

4. **CI95 and per-doc vectors.** Few prior KIE benchmarks publish bootstrap CI95 or per-doc score vectors; the framework's publish-gate (resamples 2000, seed 20260516) requires them, which is why even the attested SAVIOR cells today carry `ci95=null` until per-doc vectors are recovered [1].

5. **Contamination posture.** Public KIE corpora (FUNSD, CORD, SROIE, DocILE, Kleister, DeepForm) are widely indexed and likely in modern VLM pretraining mixes; analyses such as [24] document information-redundancy and lexical-bias problems that inflate scores. The framework's response is a private held-out rubric plus an explicit factory-rerun green-gate.

6. **Held-out / SDK coverage.** `parse-intent` is currently outside the pinned SDK surface for live realqa (PENDING-SDK) [11]; numbers exist only offline, a transparency caveat that public benchmarks are not required to disclose.

D-LAYOUT: layout-element accuracy + reading-order (PaIRS)

D-LAYOUT scores whether a parse engine recovers a document's spatial organization (the tree of titles, paragraphs, tables, figures and captions, together with the *order* in which a human would read them), rather than merely the characters those regions contain. The Parse-x framework operationalises this as three metrics: `layout_element_accuracy` (element-match F1 over the gold layout tree), `reading_order_accuracy` (Spearman footrule between the predicted and ground-truth element orderings; alternative metrics in the field include Kendall tau, the Reading Edit Distance Score (REDS), and the Layout Ordering Error Rate (LOER) of Coquenet et al. [197]; see [196] for a recent EMNLP-2024 treatment of reading-order modelling), and `pairs_layout` (PaIRS, a prompt-controlled pairwise z-euclidean spatial-relationship score, vendored from `hyprbots/vlm_ocr@1fbbc334`) [25]. The dimension explicitly addresses SAVIOR-Bench Recommendation R8, which flagged that no layout-awareness ground-truth was available in the original SAVIOR corpus and listed the facet as "defined, not yet measured" [1].

Prior benchmarks

PubLayNet (IBM, ICDAR 2019) is the de-facto pre-training corpus for document-layout detection: ~360k PubMed Central pages with auto-generated bbox annotations for five classes (text, title, list, table, figure) scored by COCO-style mAP [26]. Its scale is unmatched; however, its label set is narrow and its scientific-article distribution is monolingual English and stylistically homogeneous. it has no reading-order annotation and no element-tree.

DocLayNet (IBM, KDD 2022) hand-annotated 80,863 pages spanning six categories (financial reports, scientific articles, laws, manuals, patents, government tenders) with 11 layout classes; human baselines sit roughly 10 mAP above the strongest detectors at release [27]. DocLayNet adds class breadth and human-IAA discipline but, like PubLayNet, is a *detection* benchmark; it scores bboxes, not reading order, and does not score table or figure internal structure.

M6Doc (SCUT, CVPR 2023) introduces 9,080 modern document pages with **74 fine-grained label types** and 237,116 instances, covering scanned, photographed and PDF formats across textbooks, magazines, newspapers and test papers in Chinese and English [28]. M6Doc is, to our knowledge, the most label-rich layout corpus to date; however, it does not annotate reading order, and its 9,080-page count positions it as an evaluation set rather than a pre-training corpus. **UDOP** (Tang et al., CVPR 2023) [156] is a foundational precedent for unified vision-text-layout modelling: it casts detection, parsing, KIE and QA into a single seq2seq encoder-decoder over text + image + layout tokens, and is the most direct prior-art analogue to the joint element + order + spatial-fidelity scoring posture D-LAYOUT adopts.

ReadingBank (Microsoft, paired with LayoutReader, EMNLP 2021) is the canonical reading-order benchmark introduced by Wang et al. [207]: ~500k pages where order is harvested as weak supervision from Word-XML and then reprojected onto the rendered PDF; metrics are BLEU / token-rank Average Page-level Accuracy. LayoutReader achieves near-ceiling on the in-distribution split [29]. A limitation of

ReadingBank is that order labels inherit Word's linear flow; consequently, multi-column scientific or financial layouts may be undersupported.

OmniDocBench (OpenDataLab, CVPR 2025) is the closest contemporary to Parse-x: 1,355–1,651 PDF pages, 10 document types, 5 layout types, 5 languages, 100k+ region annotations with reading order, evaluated by Normalised Edit Distance (Normalized Edit Distance (NED)) over the predicted reading sequence [30]. v1.5 adds layout-type-stratified NED splits. OmniDocBench measures end-to-end reading order via NED but does not isolate `pairs_layout`-style PC-vs-NPC comparability, and its taxonomy does not align 1:1 with Parse-x's 16-doc-type taxonomy.

SAVIOR-Bench layout facet is defined (element accuracy, reading-order accuracy, PaIRS) but explicitly *not measured* in the original release; the corpus carries 509 docs with extraction gold and an IAA-attested 0.761 fuzzy-string agreement on 187 double-annotated pairs, but no layout-tree gold [1].

Models surveyed

Layout-aware document parsing in 2025-2026 stratifies into four families.

- **Pure layout detectors** trained on PubLayNet/DocLayNet/M6Doc: DiT, LayoutLMv3, VGT (Vision Grid Transformer) [31], YOLO-doc derivatives, and detectors pretrained on the 500K-page DocBank weak-supervision corpus [206]. Output comprises bbox plus class, with neither reading order nor transcription.
- **Layout-aware document VLMs** that emit structured markup with positional or order tags: **PaddleOCR-VL-0.9B / VL-1.5** use a two-stage pipeline where **PP-DocLayoutV2** detects regions *and* predicts reading order before the 0.9B VLM transcribes [32]. In Parse-x measurements, PaddleOCR-VL-1.5 holds the #1 layout-proxy slot at PaIRS 0.8766 on a zero-shot SAVIOR run [1].
- **Native-bbox OCR-2 models**: **Chandra** (`datalab-to/chandra-ocr-2`) is the only in-scope engine emitting bbox-grounded HTML natively, reaching mean PaIRS 0.6472 on the ParseBench n=10 table split (see App. A.2 canonical entry for architecture, variance range, and contamination posture) [10][25].
- **Reading-order specialists**: **LayoutReader** (seq2seq over text+layout, near-ceiling on ReadingBank) [29] and the recent **FocalOrder** (focal preference optimization for reading-order detection, 2026) [33].
- **Flat-OCR baselines**: HyperAPI parse (Paddle backbone) and Tesseract emit no spatial structure; HyperAPI parse scores PaIRS 0.2907, structurally expected for a primitive that returns flat markdown [10].

Gaps relative to the framework

Three deltas distinguish Parse-x's D-LAYOUT from the prior work above.

1. **Joint element + order + spatial-fidelity scoring.** PubLayNet/DocLayNet/M6Doc score bboxes; ReadingBank scores order; OmniDocBench scores order via NED but not paired-spatial fidelity. Parse-x appears, to the authors' knowledge, to be among the first frameworks jointly scoring `layout_element_accuracy` (tree F1), `reading_order_accuracy` (rank-correlation) and PaIRS (z-euclidean pairwise spatial; an internal Hyperbots metric vendored from hyprbots/vlm_ocr at commit 1fbbc334 and not externally validated) on a single gold corpus, namely ParseBench's `layout` split (16,325 rules over 500 PDFs) plus the Inv3D upstream share (~46 docs) of SAVIOR [25].

2. **PC-vs-NPC discipline on PaIRS.** SAVIOR §2.2 explicitly retracts any PC-vs-NPC PaIRS contrast [1]; Parse-x carries the same caveat as a publish-gate condition. No other public benchmark surveyed in this review flags this comparability boundary.

3. **Cross-mapping to a 16-doc-type taxonomy.** OmniDocBench's 10 doc-types and DocLayNet's 6 categories are coarser than Parse-x's 16-type taxonomy, and none route layout scores back to a downstream parse-intent F1 (D-DOWNSTREAM) for the same document. The framework's published-snapshot is therefore one PaIRS-zero-shot leader (PaddleOCR-VL-1.5 at 0.8766) and *zero* `layout_element_accuracy` / `reading_order_accuracy` cells; a stub that the ParseBench port (BLK-14) is scheduled to fill [25].

D-BBOX: bounding-box IoU + field-to-region grounding

Relation to D-LAYOUT: D-BBOX is best understood as a granularity sub-axis of D-LAYOUT (token vs. element vs. field-region bounding boxes) rather than a fully orthogonal dimension; the two share the underlying spatial-recovery surface, and a system performing poorly on D-LAYOUT element-bbox F1 will, in general, also perform poorly on D-BBOX field-region IoU.

D-BBOX scores whether a parse engine can *localize* what it reads by emitting a bounding box for each token, semantic element or extracted field and, under the field-to-region variant, mapping each field in the structured output back to its originating page region. The framework pins two metrics: `bbox_iou` (mean IoU of predicted vs. gold boxes over matched tokens) and `field_to_region_grounding_accuracy` (fraction of extracted fields whose predicted box overlaps the gold token-region at IoU ≥ 0.5) [34]. The dimension is consequential because downstream legal, audit and review workflows require citation back to the page; a parser that emits well-formed HTML without geometry therefore fails the requirement most relevant to production use [34]. This is also the dimension for which prior art is thinnest. SAVIOR-Bench explicitly logs `bbox_iou` and `field_to_region_grounding_accuracy` as **defined-but-not-measured** under risk item R9, because no gold field-box corpus was in scope [35].

Prior benchmarks

- **DocLayNet**: 80,863 human-annotated PDF pages with bounding boxes across 11 layout classes (text, title, table, figure, list, page-header, page-footer, etc.), released in COCO format and scored with mAP@IoU[0.50:0.95] [27]. Reported baselines include Mask R-CNN ResNet-101 at 73.5 mAP, Faster R-CNN at 73.4 and LayoutLMv3 at 75.1, each approximately 10 points below inter-annotator agreement [27]. **What it does not measure relative to our framework:** layout-element boxes only, no field-to-region grounding (no notion of "extract `invoice_total` → which box did it come from"); single (mostly English) corpus, low table-density.
- **PubLayNet**: 358,353 biomedical PDF pages auto-annotated from PMC XML with bounding boxes for {text, title, list, figure, table}; mAP@IoU[0.50:0.95] scoring [26]. **Does not measure:** narrow domain (PMC research articles), boxes are auto-aligned from XML and known to drift, no field-level grounding, no handwriting/forms.
- **FUNSD** [9] (see App. B.1): supplies word-level boxes and intra-document Q->A entity links. **Does not measure:** tiny scale; single-page English forms; entity-link grounding is intra-document, not field-to-region under a downstream-extraction schema.
- **Inv3D**: 25,000 synthetic 3D-warped invoice samples (plus `Inv3DReal` of 360 real smartphone captures) released with per-token boxes and template-anchored ground truth, originally built for document unwarping [36]. SAVIOR-Bench's bbox-eligible slice (~9.04%, ~46 docs) draws from Inv3D's upstream boxes [35]. **Does not measure:** synthetic-dominated, invoice-only domain, no extraction-task field schema layered on top.

- **ICDAR 2023 Robust Layout Segmentation Competition (DocLayNet challenge)**: competition track using DocLayNet's COCO format with mAP@IoU[0.50:0.95] over 11 classes [37]. **Does not measure:** layout segmentation in isolation; no end-to-end "parse-then-localize-the-field" task.
- **SAVIOR-Bench R9 ("bbox grounding")**: defines `bbox_iou` and `field_to_region_grounding_accuracy` over a planned financial-document corpus but logs them as *not yet measured*; flagged as the single largest unmeasured surface in the benchmark [35].
- **Silver-bbox pipeline (this project)**: internal spec [38] auto-derives token boxes from digital PDFs via PyMuPDF with a pdfplumber 10% cross-check (IoU < 0.95 → `silver_confidence: low`); every cell is stamped `gold_kind: "silver-pymupdf"` and may never be flipped to gold without human annotation. This pipeline unblocks drift and agreement signals on digital PDFs but is explicitly **not leaderboard-publishable** for accuracy claims.

Models surveyed

- **Native bbox-emitting OCR engines. Chandra** is the in-scope reference for native bbox-grounded HTML emission, covering all four ParseBench primary metric surfaces by default (see App. A.2 canonical entry) [25][39][34]. **Surya** (`datalab-to/surya`) emits layout boxes and benchmarks on PubLayNet/DocLayNet using a coverage-based scorer (≥0.5 = match) rather than IoU; reports precision 0.782 on DocLayNet, 0.751 on BCE [40]. **KOSMOS-2.5** is a multimodal literate model that generates "spatially-aware text" (line text concatenated with bounding boxes via a special operator) for document text recognition [41].
- **Layout-detector models scored on DocLayNet/PubLayNet.** Mask R-CNN ResNet-101 (73.5 mAP), Faster R-CNN (73.4), YOLOv5 family, LayoutLMv3 (75.1 mAP with text-modality fusion) [27]. These are detection-only models and do not parse content.
- **Document-VLMs with weak/no geometry.** GPT-4o, Gemini-1.5-Pro, Qwen2-VL/Qwen3-VL, Intern-VL2, mPLUG-DocOwl1.5, GOT-OCR-2.0 (per DOSSIER §2 model lists [42]) generally do *not* emit per-token boxes in their default decoders; they pass D-OCR but fail D-BBOX structurally.
- **Plain-text OCR primitives.** Tesseract, PaddleOCR-v4/VL-1.5, and the HyperAPI parse primitive emit text without spatial geometry on the public surface; D-BBOX is structurally untestable on these without an SDK change (carry-over Q-DF-7) [34].

Gaps relative to the framework

1. **Field-to-region grounding is essentially un-benchmarked.** DocLayNet, PubLayNet, ICDAR-2023 measure *layout* boxes; FUNSD measures *entity* boxes; none score "extracted field `total_amount` → originating page region at IoU ≥ 0.5". The `field_to_region_grounding_accuracy` metric constitutes the framework-novel surface, building on SAVIOR-Bench R9's definition [35].

2. **Gold paucity for field boxes.** Inv3D + FUNSD give ~90 docs of upstream gold token boxes; ParseBench's `layout` split adds 16,325 rules across 500 PDFs but at element (not field-schema) granularity [43]. Closing R9 with gold annotations is a partner-annotation requirement; the silver-bbox pipeline closes it *partially* without additional funding [38].

3. **Apples-to-oranges scoring across prior benchmarks.** DocLayNet uses COCO mAP@[0.50:0.95]; Surya's published numbers use coverage-with-0.5-match (not IoU); FUNSD reports F1 over entity bbox+label. The framework pins a single `bbox_iou` mean over matched tokens together with the IoU ≥ 0.5 grounding rate; consequently, cross-engine cells are comparable.

4. **Native-bbox emission is rare.** Of approximately 40 v0-runnable models in the DOSSIER, only Chandra (and, partially, Surya and KOSMOS-2.5) emits geometry by default; the rest fail D-BBOX *structurally*, not numerically. The framework reports these as "structural 0" rather than as a quality gap [34].

5. **Domain coverage.** Prior corpora are biomedical (PubLayNet), corporate-mixed (DocLayNet), forms (FUNSD), or invoice-synthetic (Inv3D). The 16-doc-type taxonomy surfaces uncovered slices (handwritten notes, tax forms, statements, contracts) where no D-BBOX cell is currently runnable even with the silver pipeline.

D-TABLE: GTRM + TableRecordMatch + cell F1

Table structure recognition (TSR) constitutes the central technical challenge of enterprise document parsing: invoices, packing slips, bank statements, 10-Q financials, and insurance schedules are dominated by tabular content with spanning headers, merged cells, borderless rules, repeated multi-page headers, and locale-specific currency formatting [44]. A transcription with high word recall may nevertheless misassign rows and columns, which degrades downstream extraction [44]. D-TABLE in the Parse-x framework therefore scores table-bearing documents (T-04 bank statement, T-11 financial statement, T-16 packing slip) on whether the engine recovers both the grid topology and the cell content, using the ParseBench `table` split (n=503 ground-truth HTML tables, 284 documents) with **GTRM** (the unweighted average of GriTS and TableRecordMatch; this composite is a Parse-x convention rather than a field-standard metric, and readers may prefer to report GriTS and TableRecordMatch separately) as primary, and TEDS plus cell-F1 as secondaries [parsebench2026, framework2026].

Prior benchmarks

- **PubTabNet** (Zhong et al., IBM): ~568k scientific tables from PMC articles, distributed with HTML structure-and-content ground truth. It introduced **TEDS** (Tree-Edit-Distance-based Similarity) split into TEDS-Structure (TEDS-S) and TEDS-Cell (TEDS-C) per Zhong et al.'s original convention [208], and **TEDS-Struct**, and remains the canonical TSR pretrain anchor [45]. TEDS, and by extension the GriTS / GTRM lineage, ultimately rests on the **Zhang-Shasha** tree-edit-distance algorithm [158], which provides the $O(m^2n^2)$ ordered-labelled-tree edit cost that the framework's table scorer must port deterministically to be cross-engine comparable. Limit relative to D-TABLE: scientific-paper distribution only; no enterprise finance distribution, no bbox geometry on every cell, and TEDS is sensitive to small text noise so structure vs content errors are conflated unless paired with TEDS-Struct [44].
- **FinTabNet** (Zheng et al., IBM, WACV 2021): ~113k tables across ~70k pages from S&P-500 annual reports, with cell bboxes and structure (train/val/test = 91,596/10,635/10,656) [46]. Closest public proxy to enterprise finance tables, but the distribution is digital-PDF-rendered 10-Ks only, with no scans, no invoices, no packing slips, no multi-page table stitching across the doc-type taxonomy the framework requires.
- **PubTables-1M** (Smock et al., CVPR 2022): ~948k tables with joint detection + structure annotations; introduced the **Table Transformer (TATR)** baseline and the **GriTS** metric family (GriTS-Top topology, GriTS-Con content, GriTS-Loc IoU geometry), formally published at ICDAR 2023 [pubtables1m2022, grits2023]. GriTS operates on the 2D cell grid (not a tree) and handles non-tree spanning structures more robustly than TEDS [47]. Limit: same scientific-PDF bias as PubTabNet; no enterprise scans; GriTS-Loc requires bbox GT that most non-PubTables corpora lack.
- **ICDAR-2013 Table Competition and ICDAR-2019 cTDaR**: small, hard mixed-domain sets including historical and handwritten tables; canonical home of **adjacency-relation F1** (precision/recall over hori-

zontally/vertically adjacent cell pairs whose endpoint text and direction both match), which decomposes errors by split / merge / shift type [44]. Limit: corpus is small and not enterprise-representative; metric depends on a normalizer the framework must pin identically to pred and gold.

- **TableBank** (Li et al., LREC 2020): ~417k tables harvested from Word and LaTeX sources, primarily a *detection* benchmark with weak structure labels [44]. Useful for detection breadth; insufficient for cell-F1 / GTRM scoring.
- **RD-TableBench** (Reducto, 2025): open-source enterprise-leaning table benchmark with messy real-world scans; designed as an industry counterweight to scientific-only TSR sets [48]. Recent and small relative to FinTabNet; metric protocols vary by reporter.
- **ParseBench `table` split** (LlamaIndex / community, 2026): 503 pages across 284 documents, distributed under Apache-2.0 with per-doc HTML `expected_markdown`; uses **GTRM** = mean(GriTS, **TableRecordMatch**) where TableRecordMatch treats a table as a bag of records keyed by column header(s) and is therefore insensitive to row/column reordering but heavily penalizes transposed/dropped headers [49]. This is the gold the framework ports for D-TABLE; the split does not cover doc-type stratification (T-04 vs T-11 vs T-16 are not separable inside the 503 without re-labeling) or the Q-LOW / multilingual strata.

Models surveyed

- **Specialist TSR.** Table Transformer (TATR) on PubTables-1M [50]; earlier lineage DeepDeSRT, TableNet, SPLERGE, LGPMA, TableFormer / EDD (the PubTabNet HTML-token convention) [44]; PaddleOCR PP-Structure as a layout+TSR+OCR cascade. Camelot/Tabula are rule-based (lattice/stream), strong only on ruled digital PDFs.
- **OCR-2.0 / end-to-end structure emitters.** GOT-OCR2.0 (HTML/Markdown/LaTeX tables); Nougat (academic-PDF Markdown, brittle off-domain); MinerU / PDF-Extract-Kit pipelines that wrap specialist TSR [44].
- **Prompted VLMs (the framework's open-prompted SOTA route).** Qwen2.5-VL 3B/7B/32B/72B (HTML/Markdown table emission with bbox-grounded cells); InternVL 2.5 / 3; dots.ocr; PaddleOCR-VL; olmOCR; DocOwl/UReader; MiniCPM-V; Phi-3.5/Phi-4-vision and the 5.6B Phi-4-multimodal-instruct (text+vision+speech; OCRBench 84.4, DocVQA 93.2) [201] [44]. Chandra is included for table extraction with documented per-doc PaIRS variance on the ParseBench n=10 table sub-slice (see App. A.2 canonical entry) [25].
- **Closed frontier (anchor-only, not products).** GPT-4o/4.1, Gemini 1.5/2.x, Claude 3.x, used as scientific anchors, never as winners [44].

Gaps relative to the framework

1. **GTRM is a recently proposed composite and has not yet been incorporated into public scorer implementations.** GriTS and TableRecordMatch each exist [grits2023, parsebench2026], but their unweighted average is a ParseBench-internal convention; the framework must port the implementation deterministically (Zhang–Shasha for TEDS, factored row/col alignment + Hungarian for GriTS, identical lenient HTML normalizer applied to pred and gold) before any GTRM cell is publishable [tabular-survey2026, framework2026]. As of the framework snapshot, **zero GTRM cells have been published**; TR-000 is REGISTERED-NOT-YET-RUN [25].

2. **Doc-type stratification.** No prior table benchmark cleanly separates bank statement vs 10-Q vs packing slip vs insurance schedule; the framework's 16-doc-type taxonomy crossed with D-TABLE exposes per-doc-type collapse modes that PubTables-1M / FinTabNet aggregate away.

3. **Quality and language strata.** PubTabNet/FinTabNet/PubTables-1M are digital-PDF clean; ICDAR-2019 cTDaR is the only hard-scan anchor and is small. D-TABLE × D-DOC-QUALITY (Q-LOW JPEG-Q40 / rotated / blurred re-renders) and × D-LANG overlays are framework-native crosses with **no public corpus** today.

4. **Proxy-metric honesty.** The current Azure DI long-invoice cell carries line-count preservation 0.6396 / line-item set-F1 0.3439 (proxies, not GTRM) and the framework explicitly marks them as such rather than back-fitting to GTRM [25]. Prior leaderboards have substituted proxies without disclosure.

5. **Apples-to-apples decoding pins.** ParseBench-reported numbers mix zero-shot vs fine-tuned VLMs, varying TEDS variants, and unpinned decoding [44]; the framework's pinned-prompt + decoding contract is the missing comparability layer.

2.5.1 Survey landscape (table recognition + VLM-era document AI)

Four surveys published between 2024 and 2025 frame the literature within which D-TABLE and the broader Parse-x dimensions are situated; naming them explicitly allows this review to state its delta rather than re-derive their taxonomies.

Shi and Shi [167] ("A Review On Table Recognition Based On Deep Learning", arXiv:2312.04808) organize TSR into datasets/metrics, two-stage detection+structure models, end-to-end models, and data-centric augmentation. They consolidate PubTabNet, FinTabNet, ICDAR-cTDaR, PubTables-1M, and the TEDS/GriTS metric family that D-TABLE depends on, but they do not cover enterprise stratification (bank statement vs 10-Q vs packing slip), the Q-LOW degraded-scan overlay, or the GTRM = mean(GriTS, TableRecordMatch) convention that ParseBench introduces; the review is also distributed in Chinese with a GitHub companion, which limits direct reuse.

Ding et al. [168] ("A Survey on MLLM-based Visually Rich Document Understanding", arXiv:2507.09861, ACL 2026 Findings) provides the closest match to the VLM-era component of Parse-x. It taxonomizes OCR-based vs OCR-free MLLM pipelines, pretraining and instruction-tuning recipes, multi-page and multilingual handling, and the emerging RAG/agentic layer. It does not commit to a per-dimension scoring contract (no D-OCR/D-TABLE/D-BBOX separation), does not pin decoding/prompt protocols, and reports no cost-latency dimension; D-COST-LATENCY and the pinned-prompt comparability layer are therefore outside its scope. The earlier Ding et al. 2024 survey [169] (arXiv:2408.01287) covers the pre-MLLM Visually Rich Document (VRD) lineage (LayoutLM family, multimodal fusion, pretraining objectives) and is the natural companion citation for §D-LAYOUT and §D-PARSE-INTENT.

Barboule, Piwowarski and Chabot [171] (arXiv:2501.02235) survey question-answering over visually rich documents and explicitly note that the field "still lacks consensus on several key aspects of the processing pipeline", which is the gap D-DOWNSTREAM tries to close by anchoring to extractive QA lift rather than free-form generation. Somvanshi et al. [170] (arXiv:2410.12034) survey deep tabular learning (TabNet, SAINT, FT-Transformer, TabPFN, diffusion-based synthesis); it is adjacent rather than central, because Parse-x scores table extraction from images and PDFs rather than downstream supervised learning on already-structured tables; nevertheless, it bounds expected performance once a parser has produced a clean grid.

Net delta of this review against the survey lineage: per-dimension scoring contracts, doc-type x quality x language stratification, pinned decoding, and a cost-latency dimension, none of which the four surveys above operationalize.

D-DOC-QUALITY: stratification by document quality (Q-HIGH/MED/LOW)

D-DOC-QUALITY addresses a question that headline accuracy figures on parse benchmarks almost always obscure: when the input is a degraded scan (JPEG-Q40 compression, ≥5° rotation, motion or defocus blur, low DPI), does the engine's accuracy collapse, drift, or remain stable? In the Parse-x framework it is explicitly a **stratification** of D-OCR and D-PARSE-INTENT rather than an independent score, with the primary metric being the signed delta in `savior_word_recall` between Q-HIGH and Q-LOW renderings of the same corpus under identical decoding pins, and a Q-MED midpoint reported for monotonicity diagnosis [51]. The degradation recipe is borrowed from [52] §10 R4. Failure modes F1 (vertical/rotated text), F5 (degraded scan) and F8 (stylized text) are expected to activate disproportionately in Q-LOW [51].

Prior benchmarks

- **SAVIOR-Bench v1**: 509 documents drawn from UCSF (47%), synthetic (18%), financial (17%), Inv3D (9%) and FUNSD (9%), with IAA 0.761 over 187 double-annotated pairs. The corpus deliberately includes "low-quality PDFs" and "degraded scans" as a class label, however, the published results are aggregate-only, with no per-quality slice, no degradation track, and no rendered Q-HIGH/Q-LOW counterparts of the same document. The authors flag this themselves as recommendation R4 ("No degradation / robustness tracks ... not done, no degraded corpus exists") [52].
- **DocIQ / DIQA-5000**: a subjective document-image-quality dataset that partitions 5,000 corrupted images into five degradation classes (luminance, distortion, blurriness, noise, compression). It is a quality-*assessment* dataset (predict the MOS), not a parse-accuracy benchmark; it does not score OCR or layout downstream of the degradation [53].
- **DocUNet**: 130 mobile-camera captures of 65 paper documents (receipts, letters, fliers, magazines, books) annotated with the flat-scan ground truth and split into "easy" (single crease/fold) and "hard" (heavy wrinkles) cases. It scores geometric dewarping (MS-SSIM, Local Distortion), not parse fidelity, but is the closest public source of paired clean/degraded scans of the same content [54].
- **RVL-CDIP**: 400,000 grayscale ~100 DPI documents from 1980s–90s scanners, which are *intrinsically* low-quality but not stratified: there is no Q-HIGH counterpart of the same document. Subsequent audits also report 8.1% label noise (range 1.6–16.9% by class), which contaminates any quality-versus-accuracy slope estimated on top of it [55][56].
- **Historical-OCR benchmarks (IMPACT, olmOCR-Bench historical split)**: IMPACT supplies degraded historical print as a single domain; olmOCR reports 82.3% on old math scans dropping to 47.7% on general historical scans, demonstrating the degradation gap but, again, with no controlled stratification of the same source at multiple quality tiers [7][57].
- **Augraphy**: a data-augmentation *library* (ink bleed, jitter, lighting, scanner artefacts) widely used to synthesise degraded training data; it provides the operators that a Q-LOW renderer needs but is not itself a benchmark [58].
- **SOULSHINE / bol-grn (Hyperbots-internal)**: 22 BoL PDFs (soulshine1) + 22 byte-identical bol-grn-test inputs (soulshine2); native mixed-quality PNG-derived scans are present (two long-tail outliers at ~140 and ~56 pages, customers omitted) but no gold and no quality stratification, only OCR-sweep counts (Tesseract 22/22, PaddleOCR 15/22) [59].

Models surveyed

Models likely to be measured under D-DOC-QUALITY are drawn from the same families already measured on D-OCR / D-PARSE-INTENT in the project dossier [42]:

- **OSS document-parse VLMs**: Chandra (`datalab-to/chandra-ocr-2`), Qwen3.6-35B-A3B, Qwen2.5-VL-3B/7B/72B, Qwen3-VL-32B/235B, MinerU-2.5 / MinerU-2.5-Pro-2604, PaddleOCR-VL-1.5, Surya, Marker-1.0, Nougat-0.1.0, mPLUG-DocOwl 1.5 / DocOwl2, GOT-OCR-2.0, FireRed-OCR, DeepSeek-OCR-2, Logics-Parsing-v2.
- **Closed VLMs accessible via API**: GPT-4o, GPT-4.1, Gemini 1.5/2.5 Flash and Pro, Claude Sonnet 4.6, Mistral-Large-3-675B, Llama-3.2-90B-Vision; routed via Azure OpenAI for the OpenAI family per project preference.
- **Legacy/baseline OCR**: Tesseract-OCR, PaddleOCR-v4, EasyOCR, OpenOCR, Mathpix, HyperAPI parse (paddle-backed).
- **Document-image-quality assessment heads (auxiliary)**: DocIQ feature-fusion network [53] and the no-reference text-line DIQA model [60]; not parsers, but candidates for predicting the Q-stratum a document will land in pre-flight.

Prior measurements on these models almost never report per-quality slices; the F-B study panel (Chandra/Qwen3.6/Tesseract/HyperAPI-parse on ParseBench n=10) is aggregate only [42].

Gaps relative to the framework

1. **No public benchmark renders the same document at controlled Q tiers.** SAVIOR-Bench labels low-quality as a *class* but does not pair it with a high-quality counterpart of the same document; DocUNet pairs warped vs flat but on a geometric axis, not on JPEG/blur/DPI; RVL-CDIP and IMPACT are intrinsically low-quality with no clean twin [52][54][55][57]. The Parse-x publish-gate of "Q-HIGH and Q-LOW counterparts of the same dataset, same model, same prompt, same decoding" therefore has no off-the-shelf source and must be built; the spec proposes synthetic re-renders of SAVIOR-Bench with frozen ImageMagick/Pillow degradation seeds [51].

2. **No signed-delta metric is standard.** Robustness studies of VLMs under common corruptions report aggregated accuracy drops [61][62] but do not pin the metric to a parse-specific recall (e.g. `savior_word_recall`) nor publish per-document deltas that would let a downstream consumer flag fragile engines.

3. **Failure-mode incidence is unreported.** F1/F5/F8 incidences in the Q-LOW stratum are explicitly called out by Parse-x; no prior benchmark publishes per-failure-mode F-scores stratified by quality [52][51].

4. **Internal datasets are not yet stratified either.** The SOULSHINE BoL corpus is native mixed-quality but has no gold and no Q-tier labels [59]; today's snapshot is "zero Q-* stratified cells" [51].

5. **Quality-assessment heads are not coupled to parse benchmarks.** DocIQ and DIQA-5000 predict subjective quality but are not run as gates upstream of a parse leaderboard [53][60]; Parse-x's Q-tier assignment is currently rule-based (JPEG-Q40/rotation/blur/DPI), leaving an open question of whether learned IQA scores should replace the rules.

Limited prior art segments parse accuracy by quality tier on the *same content*; this constitutes the principal contribution opportunity for the D-DOC-QUALITY dimension.

D-LANG: multilingual evaluation overlays (EN/DE/FR/ES/ZH/JA/AR)

D-LANG addresses a simple but operationally critical question: does an engine's transcription and parse quality remain stable when the *same* document template is rendered in a non-English language or script? The Parse-x framework spec defines this dimension as a **stratification overlay**, not a standalone task; the primary metric is the *signed delta* in `savior_word_recall` (D-OCR) and field-level parse accuracy (D-PARSE-INTENT) between an L-EN baseline and each of L-DE/FR/ES/ZH/JA/AR strata [51]. This framing is motivated by SAVIOR-Bench's Pillar-6 finding that a "multilingual / very-dense-layout" residual cluster persists even after fine-tuning, that is, the gap persists and is not closed by SAVIOR-Train [1]. The framework's gold-construction recipe (rendering translated text into the same LibreOffice template, plus native partner annotation for L-ZH/JA/AR) is explicitly designed to hold layout constant so any delta is attributable to script and language, not template drift [51].

Prior benchmarks

- **MTVQA** (ByteDance, 2024; ACL Findings 2025): 6,778 human-annotated QA pairs over 2,116 text-centric images across 9 low-resource languages (AR, DE, FR, IT, JA, KO, RU, TH, VI). Reports GPT-4o/4V/Claude-3/Gemini all "with large room for improvement"; per-language accuracy is the primary metric [63]. **Does not cover**: ZH, ES; no layout-held-constant control; no separation of OCR error from VQA-reasoning error, so cannot be used directly as a D-OCR overlay.
- **XFUND** (Microsoft, ACL Findings 2022): 1,393 forms in 7 languages (ZH, JA, ES, FR, IT, DE, PT), 199 per language, with SER and RE tasks; extension of the English FUNSD [64]. **Does not cover**: AR, EN (defers to FUNSD), and provides only form-style documents, narrow over the Parse-x 16-doc-type taxonomy. Metric is entity-level F1, not transcription-level edit distance, so it does not score D-OCR directly. The companion **LayoutXLM** model [155] extends LayoutLMv2 with multilingual pre-training over 53 languages and is the canonical encoder evaluated on XFUND; it remains a key multilingual document-understanding baseline against which D-LANG-overlay deltas can be referenced.
- **OCRBench v2** (Fu et al., arXiv 2501.00321): bilingual (EN/ZH) benchmark of 10,000 human-verified QA pairs across 23 tasks and 31 scenarios, with a 1,500-image private split [3]. **Does not cover**: any of DE/FR/ES/JA/AR; the EN/ZH split is the binary axis, not the 7-language overlay Parse-x requires.
- **MDPBench** (cited in OCRBench v2 ecosystem): 3,400 document images across 17 languages including Simplified/Traditional Chinese, EN, AR, DE, ES, FR, JA, the closest existing match to the Parse-x 7-language target set [65]. **Does not cover**: layout-controlled rendering of the same template across languages, so per-language deltas conflate template variance with language variance.
- **M3DocVQA / M3DocRAG** (Cho et al., arXiv 2411.04952, ICCV 2025 Workshop): 3,000+ PDFs, 40,000+ pages for multi-page multi-document VQA [66]. **Does not cover**: language stratification is incidental rather than designed; primarily an EN long-document retrieval benchmark.
- **M4-ViteVQA** (NeurIPS D&B 2022): 7,620 video clips, 25,123 QA pairs over 9 categories for video scene-text VQA [67]. **Does not cover**: documents (it is video-scene-text), and is largely monolingual; useful only as a reference for video-OCR multilingual extensions, not D-LANG itself.
- **Language-specific OCR corpora**: SCUT-EPT, 50,000 handwritten Chinese exam-paper text lines, 4,250 character classes, the standard Chinese HCTR benchmark [68]; MTWI, Chinese web-image scene text (2018 ICPR competition track, ~20k images) [69]; and **synthetic Japanese** lines used to bootstrap PaddleOCR's `japan` model [70]. These provide per-script transcription gold but are not template-aligned to any English baseline.

Models surveyed

Per DOSSIER §2, the v0-runnable OCR/VLM panel includes both models with explicit multilingual coverage and models that are English-only by default, a contrast the overlay is designed to expose:

- **Multilingual-by-design**: **PaddleOCR-VL-1.5** and **PaddleOCR-v4** (CN/EN/JP/KR plus 80+ alphabet languages via `--lang` flag) [70]; **Qwen2-VL-7B/72B** and **Qwen3-VL-235B** (Alibaba, multilingual text+vision pretraining covering ZH/EN/JA/KO/DE/FR/ES/AR) [71]; **InternVL2-8B/26B/76B** (multilingual pretraining including ZH/EN) [72]; **GLM-OCR** and **MinerU-2.5** (Chinese-first OSS); **GPT-4o**, **Gemini-1.5-Flash/Pro** (broad multilingual VQA, evaluated on MTVQA) [63]; **Mathpix** and **Textract** (multilingual mode by configuration).
- **English-first / multilingual-by-pack**: **Tesseract-OCR** ships English by default; multilingual support requires the `tesseract-lang` data pack (165 traineddata files, 685.7 MB) which the Parse-x harness only installed on 2026-06-10, before which Chinese inputs threw `UnicodeDecodeError` and Tesseract `text_content` scored 0.0 on the diverse n=10 ParseBench split [42]. This is exactly the structural failure mode D-LANG is meant to surface.
- **Latin-script-leaning**: **Marker-1.0**, **Nougat-0.1.0**, **GOT-OCR-2.0**, strong on EN academic PDFs, with limited or undocumented L-ZH/JA/AR coverage [42].

Gaps relative to the framework

1. **No prior benchmark holds layout constant across the L-EN→L-* axis.** XFUND, MTVQA and MDPBench all vary template and language together; Parse-x's LibreOffice-rendered translation overlay is the first to isolate language as a single-variable delta on a fixed template [51].

2. **Per-language D-OCR vs D-PARSE-INTENT separation is absent in prior art.** MTVQA, M3-DocVQA and OCRBench v2 score end-to-end answer accuracy, conflating recognition error with reasoning error. Parse-x decomposes the delta into D-OCR (word recall) and D-PARSE-INTENT (field-level) per language stratum.

3. **SAVIOR's F9 "multilingual residual" has no per-language counts**; the SAVIOR PDF explicitly states "no per-class counts available" for the multilingual residual cluster [1]. Parse-x's 7-language stratified report directly fills this measurement hole.

4. **English-only-by-default tooling is invisible to prior benchmarks** that ship a multilingual harness. Parse-x logs the Tesseract-without-tesseract-lang failure as a first-class result, not a setup bug, surfacing a procurement-relevant gap [42].

5. **L-AR (RTL script) coverage is thin everywhere.** Among prior benchmarks only MTVQA and MDPBench include Arabic; ZH/JA/AR partner annotation is gated in Parse-x v0 (Q-DF-5) [51], an honest scope limitation rather than a coverage claim.

D-DOWNSTREAM: parse → downstream extract F1 (parse-lift; new CEO-required dimension)

D-DOWNSTREAM scores a parse engine not by the readability of its output but by its utility as input to the next stage. Concretely, a frozen field-extractor (Qwen 3.6-35B-A3B in v0, pinned at `T=0.1, top_p=0.9, max_tokens=2048, seed=20260516`) is run twice per document: once conditioned on `parse(doc) → text`, once on the raw page image. The primary metric is `downstream_extract_micro_f1`; the headline is `parse_lift = downstream_extract_micro_f1 − no_parse_extract_micro_f1`, with paired-bootstrap CI95 over docs in the spirit of [73]. Positive lift indicates that parse contributes measurable value to the pipeline; zero or negative lift indicates that the parse stage adds no value over a direct vision-LLM baseline. The dimension has not, to the authors' knowledge, been previously formalized in parse benchmarks; virtually all prior parse leaderboards score the parse string intrinsically (edit distance, BLEU-on-markdown, layout F1, table TEDS) and infer downstream utility only by analogy. D-DOWNSTREAM addresses that inference gap by measuring it directly. Per the framework spec (lines 140–172), today's snapshot is zero published cells; harness wiring for `extractor_pin` and the paired-lift computation is on the F-A phase list.

Prior benchmarks

Prior art splits into three buckets, none of which is a clean parse-lift benchmark.

Intrinsic KIE leaderboards. SROIE (626 train / 347 test receipts, four entity fields) and CORD (1,000 receipts, 30 labels) are the canonical end-task KIE sets; both report field-level F1 against gold but condition the extractor on *gold or dataset-provided OCR boxes*, not on a swappable parse engine [74][75]. LayoutLMv3-large reaches 97.46 F1 on CORD (Table 1, official 800/100 split) [76], and the original LayoutLM-large reports 95.24 F1 on SROIE receipt understanding [76b]. These measure extractor quality given fixed text; they do not vary the parser nor compute a parse-vs-no-parse lift. FUNSD (149/50 forms) [77] is analogous for form understanding.

Document VQA / end-task substitutes. DocVQA [78] (see App. B.1; ANLS = Average Normalized Levenshtein Similarity per Mathew et al. [209]) is sometimes used as a proxy for "did parse help" (OCR-pipeline answers vs direct VQA); the Agentic Document Extraction 99.16 result outperforming OCR pipelines by 3.2 pp GT-in-Pred [79] is exactly the "parse may be net-negative" signal D-DOWNSTREAM is built to surface.

OCR-quality-impact studies. van Strien et al. [80] assessed six downstream NLP tasks (NER, dependency parsing, IR, topic modelling, LM fine-tuning, sentence segmentation) as a function of OCR character-error rate on historical newspapers, finding consistent task-level degradation and recommending ≥90% OCR quality. Chiron et al. [81] showed retrieval MAP drops materially with OCR error on a 22k-document Portuguese IR collection. Both *demonstrate* that intrinsic OCR scores predict downstream loss weakly, but neither defines a paired-lift metric nor pins a single extractor across engines.

Composite parsing benchmarks. SAVIOR-Bench [1] defines `parse = extraction + layout + bbox`; its Facet-1 extraction sub-score (parse-intent F1 / WER, in-house leaderboard; SAVIOR-Train fine-tune at F1 0.8328 / WER 0.2589; GPT-4o zero-shot 0.8052 / 0.4232) is the closest sibling, but it scores the parser's *own* extraction head; there is no frozen-downstream-extractor pin, so engines with strong native heads (e.g. layout-aware VLMs) and engines that emit only markdown text (e.g. Tesseract) are not compared apples-to-apples on downstream utility. OmniParser [23] unifies text spotting, KIE, and table recognition into one model but again co-trains parse and extract; OmniDocBench [4] scores parse intrinsically across doc types.

In summary, **limited prior art exists on paired parse-lift with a frozen extractor.** ASR has a long tradition of WER vs downstream-NLU evaluations and MT has paired-bootstrap significance since [73], but the parse-then-extract pipeline has not standardised the analog.

Models surveyed

In-scope as **parse engines** (varied across the D-DOWNSTREAM cell row): HyperAPI parse (Hyperbots production primitive), Chandra (layout+bbox+table), MinerU 2.5 / MinerU2.5-Pro-2604-1.2B, PaddleOCR-VL-1.5, Marker-1.0, Surya, DocOwl2 / mPLUG-DocOwl1.5, Nougat-0.1.0, GOT-OCR-2.0, DeepSeek-OCR-2, Logics-Parsing-v2, Tesseract-OCR (deterministic-text baseline), per DOSSIER §2.x model rosters [10].

In-scope as **frozen extractor pins** (one chosen, all engines share it): the v0 pin is Qwen 3.6-35B-A3B (Hyperbots IP fine-tune, served at `http://135.233.113.234:6006/v1`; DH-001 FieldRecall 0.4099 at n=95, CI95 [0.3454, 0.4784], 2026-06-10) [10]. Candidate alternate pins for sensitivity checks: Qwen3-VL-32B-Instruct, GPT-4.1 / GPT-5.4 thinking via Azure OpenAI, Claude Sonnet 4.6, Gemini 2.5 Pro, Mistral-Large-3-675B, Llama-3.2-90B-Vision; however, any change to the pin requires every cell to be re-run.

Gaps relative to the framework

1. **No prior benchmark publishes `parse_lift` with paired-bootstrap CI.** SROIE/CORD/FUNSD (see App. B.1) vary the extractor while holding text fixed; D-DOWNSTREAM does the inverse (frozen extractor, varied parser) and reports the *paired* difference per doc per [73]. SAVIOR-Bench reports per-engine extraction but not a swapped-extractor lift.

2. **No prior benchmark reports per-doc-type lift across a 16-type taxonomy.** Aggregate F1 hides that parse may lift on T-07 bank-statement (long tabular pages) and degrade on T-08 cheque (short image where direct VQA wins); the framework requires a 16-row table per engine and a `weighted_parse_lift` aggregate.

3. **Extractor-leakage is unaddressed in prior pipelines.** Because no prior public benchmark pins the extractor, the question "is the lift you measured a property of parse or of an extractor trained on these docs?" is unanswerable. The framework names the risk explicitly (Q-DF-2; Qwen 3.6 is a Hyperbots IP fine-tune; contamination audit is a follow-on item).

4. **Image-only doc handling.** OCR-impact studies [80] hold text-OCR as the only input; D-DOWNSTREAM specifies that the baseline leg for image-only docs (T-08 cheque PNG) is raw image to the same extractor, so the no-parse baseline is well-defined for every doc type.

5. **Engine-error honesty.** Long-doc 300s timeouts in the extract leg surface as `engine-error` cells rather than silent skips (BLK-4), preserving the integrity of the lift mean.

D-COST-LATENCY: $/correct and p50/p95/p99 latency

The D-COST-LATENCY dimension addresses whether a parse engine is deployable, not merely accurate. The Parse-x framework spec defines its primary metrics as `latency_p50_ms`, `latency_p95_ms`, `latency_p99_ms`, and `dollars_per_correct_call` (cost divided by count of calls that clear the accuracy threshold of the relevant primary metric), with failure rate and throughput-at-concurrency-1 as secondary metrics [51]. The "$/correct" formulation reflects a design choice: a low-cost engine with a 40 percent failure rate does not represent low effective cost, and an engine with a high p99 timeout rate does not represent low effective latency. This dimension additionally runs as a secondary on every accuracy cell, an unusual configuration: most parse/OCR benchmarks ignore wall-clock and price entirely, and most LLM efficiency benchmarks ignore correctness.

Prior benchmarks

HELM (Stanford CRFM). HELM is the canonical "holistic" framework: each of its 16 core scenarios is scored on 7 metrics including efficiency-as-latency and efficiency-as-cost alongside accuracy, calibration, robustness, fairness, bias, and toxicity [helm2022, helmdocs2026]. Efficient-HELM further offers fractional-sample runs that preserve ranking with far fewer tokens [82]. Relative to Parse-x, HELM measures *idealized* inference cost (denoised energy/$ estimates, often per query token count rather than wall-clock under load) on text-only scenarios; it does not normalize cost against per-task correctness, and its document/vision coverage (VHELM) is shallow on parse-style tasks like table-cell or bbox extraction [83].

Artificial Analysis. Artificial Analysis is a widely used third-party leaderboard that pins all models to identical probe hardware and reports median output speed (P50 over 72 hours), time-to-first-token latency, and blended input/output price per million tokens [84]. By their March 2026 cut, the price spread across frontier models is roughly 250x bottom-to-top and YoY prices have dropped ~80% [85]. The methodology is rigorous on serving-side metrics but the workload consists of short-prompt / short-completion chat rather than multi-page PDFs with image tokens, and no parse-correctness gate is applied.

MLPerf Inference. MLCommons' MLPerf Inference v5.1 and v6.0 are the hardware-vendor standard, with audited submissions reporting Offline tokens/sec and Server p99 TTFT for models like Llama-3.1-8B and GPT-OSS-120B; v6.0 specifically tracks p99 TTFT for Mixture of Experts (MoE)-routed serving (e.g., a 2,903 → 2,001 ms reduction via the BLAZE routing patch) [mlperf51-2025, mlperfv6-2026]. MLPerf measures serving-stack efficiency under SLOs, but the "tasks" are LLM-shaped (chat, summarization), not document parse with ground-truth gold; $/correct is not a MLPerf concept.

LLM-inference latency surveys. Production-oriented metric guides (BentoML, GMI Cloud) codify the p50/p95/p99 + TTFT + ITL + throughput stack and explicitly warn that "a 0.4 s p50 / 4 s p99 looks great in demos and breaks in production" [bentomlmetrics2026, gmicloud2026]. These sources serve as methodology references rather than benchmarks.

Vendor pricing pages and OCR-cost surveys. Mistral OCR is published at $1–2 per 1,000 pages [86]; Google Document AI ranges $1.50–$30 per 1k pages by processor; AWS Textract is $1.50/1k for basic OCR and $65/1k for forms-and-tables [87]. Frontier VLM passthrough (e.g., Gemini 2.5 Pro on a scanned page ≈ $0.13/page) costs ~60–167x more than self-hosted OCR-VLM pipelines [parslicost2025, ofoxocr2026]. None of these prices is joined to a per-task correctness gate.

SOURCE-EXTRACT VLM-OCR (Hyperbots internal corpus). The on-disk SOURCE-EXTRACT benchmark reports fine-tuned-Qwen-VL inference at p50 2,548 ms / p99 3,319 ms, 3% failure, and infrastructure-derived $0.000191 per successful call (≈ $0.19 / 1k) on A100-40GB at ~$2/hr [2]. Absent from that report are p90/p999, full latency histograms, and per-model LoRA cost amortization: exactly the secondary metrics Parse-x flags as gaps.

Models surveyed

Relevant model families for this dimension fall into four tiers:

- **Frontier hosted VLMs** (Gemini 2.5 Pro, GPT-5.x via Azure OpenAI per [Hyperbots Azure OpenAI routing policy], Claude 4.x): high per-page cost (~$0.05–$0.13/page), high p50 latency on multi-image inputs, but strong accuracy [artificialanalysis2026, parslicost2025].
- **Cheap hosted OCR** (Mistral OCR 3, Google Document AI OCR, AWS Textract OCR): $1.50–$2 / 1k pages, sub-second-per-page typical, narrow output schema [mistralocr2025, docaicost2026].
- **OSS local VLM/OCR** (GLM-OCR 0.9B at 94.62 OmniDocBench, PaddleOCR-VL-1.5 at 94.50, MinerU-2.5, Chandra): negligible marginal $ if amortized, but p50 latency varies from sub-second (PaddleOCR) to ~8.5 s/page (Chandra smoke test, 528 prompt + 821 completion tokens) [ofoxocr2026, parsexframework2026].
- **Fine-tuned proprietary** (SAVIOR-Train Qwen-VL LoRA): p50 2.5 s, $0.19 / 1k successful on A100 [2].

- **Serving-stack baselines** (Llama-3.1-8B, GPT-OSS-120B per MLPerf): not parse models, but their published p99 TTFT envelopes set the floor for what an LLM-judge or downstream-extract leg can add to total wall-clock [mlperf51-2025, mlperfv6-2026].

Gaps relative to the framework

Three deltas distinguish Parse-x D-COST-LATENCY from the prior art:

1. **$/correct as a primary, not a derived metric.** HELM, Artificial Analysis, and MLPerf all report cost OR accuracy but never divide them at the cell level. Parse-x incorporates the accuracy gate per dimension (e.g., D-OCR edit-distance threshold) into the cost denominator, so a 40%-failure engine reflects that failure in the headline number [51].

2. **Full p50/p95/p99 + failure rate on every accuracy cell.** Most academic parse benchmarks (OmniDocBench, ParseBench, DocVQA) publish a single accuracy column and no wall-clock at all [88]. Parse-x requires latency as a secondary on every cell and primary on the cost-cascade ensemble (E5), making cross-cell tail-latency comparisons possible.

3. **Price-list-pinned cost in `_meta`.** Prior benchmarks either use vendor list prices (Artificial Analysis) or infrastructure cost (MLPerf) but rarely commit to a pinned snapshot for reproducibility. Parse-x records the price-list timestamp per cell, which matters in a market where YoY prices fell ~80% [85]. Limited prior art on this specific reproducibility move.

Scope limitation: Parse-x does not yet measure energy-per-correct (kWh, gCO_2e), which HELM-v2 and MLPerf Power do track; this is a known follow-on, not a claimed strength.

2.8 Handwritten document recognition (coverage extension)

Handwritten document recognition (Handwritten Text Recognition, HTR) was designated out-of-scope in v4 of this review; however, the senior critique noted that the printed-text framing leaves a measurable surface (scanned ledgers, signed forms, archival correspondence) uncovered. This subsection inventories the prior art so that a future Parse-x overlay can be aligned to it without re-discovery.

Datasets. The IAM Handwriting Database [178] remains the canonical offline English line-and-word corpus (657 writers, 1,539 scanned pages, ~13k labeled lines) and defines the per-writer split convention every modern HTR paper still uses. RIMES (French administrative letters) and the Bentham collection (19th-century English manuscript pages curated under the EU READ / tranScriptorium project) extend the corpus to multi-writer historical material; READ and the recurring ICFHR / ICDAR HTR competitions (2014, 2016, 2018, with follow-on tracks at ICDAR 2021 and 2023) supply the competition-grade gold and the Transkribus platform that hosts much of the archival evaluation pipeline. None of these corpora overlap meaningfully with the printed-document benchmarks surveyed in sections 2.1 through 2.7; writer variation, baseline skew, and ligature behaviour dominate the error budget instead of layout or table structure.

Metrics. HTR reports Character Error Rate (CER) and Word Error Rate (WER) on a per-line basis, both Levenshtein-normalised; current state-of-the-art (SOTA) performance on IAM lines is approximately 2.7 percent CER / 6.3 percent WER with language-model fusion [182]. These metrics are continuous, not field-structured, so the F1-style D-PARSE-INTENT and D-BBOX dimensions do not transfer; only D-OCR (under a different normaliser) and D-COST-LATENCY apply directly.

Modern engines. Transformer encoder-decoder architectures replaced the CNN+BLSTM+CTC stack starting with TrOCR [179], which pre-trains an image transformer and a text transformer jointly and reports SOTA on IAM, SROIE, and scene-text in a single recipe. The 2025 HTR survey [180] partitions the field into line-level and beyond-line-level (paragraph, page) recognition and notes that page-level transformer HTR is still pre-SOTA versus line-level pipelines, an open problem distinct from the printed-document layout work in section 2.3. A March 2025 benchmark of proprietary multimodal LLMs (Claude 3.5 Sonnet, GPT-4o, Gemini) against Transkribus on English / French / German / Italian handwritten lines [181] finds neither approach dominates: VLMs outperform on contemporary English and underperform on historical non-English material, and they cannot self-correct zero-shot outputs.

Delta from this framework. Handwriting restructures the failure-mode taxonomy enumerated in section 2.1: F11-DICT-LITERAL (literal-string lookup failures) becomes largely irrelevant because handwritten tokens rarely match a closed vocabulary, while F12-style segmentation errors (touching characters, broken cursive strokes, ascender / descender confusion) dominate and require writer-adaptive normalisers that the printed-text engines surveyed here do not implement. A Parse-x HTR overlay should therefore re-baseline against IAM and a READ-derived historical slice using CER, not field-F1, and treat any shared D-COST-LATENCY cell as the only directly comparable measurement.

Appendix A. Model Lineage

Scope: ~50 models surveyed in DOSSIER §2.1 (OCR family, 31 cells) and §2.2 (parse-intent family, 19 cells) [10]. Each row lists base architecture, training data (as publicly disclosed), license, version pin (commit / release tag / `PENDING-PIN`), and one primary citation. Rows confirmable only via the DOSSIER list, without an authoritative model card or paper read at table-build time, are clustered at the bottom under "Verified-via-DOSSIER-only; deep-citation PENDING" rather than fabricated.

A.1. Verified lineage

Model	Family	Base architecture	Training data	License	Version pin	Citation
Tesseract-OCR	OCR-classical	LSTM line recognizer (Tesseract 4/5)	100+ langs, Google internal + community traineddata	Apache-2.0	5.3.x (DOSSIER `BLOCKED-LANG CLEARED 2026-06-10`)	[89]
EasyOCR	OCR-classical	CRNN + CRAFT detector	Synthetic + COCO-Text + ICDAR	Apache-2.0	PENDING-PIN	[90]
PaddleOCR-v4	OCR-classical	PP-OCRv4 (DB-Net det + SVTR rec)	Internal Baidu + open Chinese corpora	Apache-2.0	v2.7 / PP-OCRv4 release	[70]
PaddleOCR-VL-1.5	OCR-VLM	PaddleOCR-VL (0.9B; ERNIE-4.5-0.3B LM + NaViT-style visual enc)	Disclosed in tech report; multilingual doc corpus	Apache-2.0	v1.5 (HF `PaddlePaddle/PaddleOCR-VL`)	[32]
OpenOCR	OCR-classical	Modular det+rec (SVTR / DBNet)	Open academic OCR corpora	Apache-2.0	PENDING-PIN	[91]
Surya						[92]

Model	Family	Base architecture	Training data	License	Version pin	Citation
	OCR-pipeline	Det/rec/layout/order ensemble (custom encoders)	Common Crawl PDFs + synthetic	GPL-3.0 / commercial dual	`datalab-to/surya` PENDING-PIN	
Marker-1.0	OCR-pipeline	Surya + heuristic post-processors → markdown	Same as Surya	GPL-3.0 / commercial dual	v1.0	[93]
MinerU-0.9	OCR-pipeline	LayoutLMv3 + PaddleOCR + UniMERNet	OpenDataLab internal	AGPL-3.0	v0.9.x	[94]
MinerU-2.5	OCR-pipeline	MinerU v2.5 (refactored; VLM-assisted)	OpenDataLab + extended	AGPL-3.0	v2.5	[95]
MinerU2.5-Pro-2604-1.2B	Parse-intent VLM	1.2B VLM trained for doc parsing	Per MinerU 2.5 tech report	AGPL-3.0 (model card)	`opendatalab/MinerU2.5-...-1.2B`	[95]
Nougat-0.1.0	OCR-VLM	Swin encoder + mBART decoder	arXiv PDFs (academic; ~8M pages)	CC-BY-NC-4.0	v0.1.0	[96]
GOT-OCR-2.0	OCR-VLM	Vary-style 580M encoder + Qwen 0.5B decoder	Synthetic + scene + doc OCR (200M+ pairs)	Apache-2.0	`stepfun-ai/GOT-OCR2_0`	[97]
Vary-toy	OCR-VLM	Qwen-1.8B + new vision vocabulary	Vary doc/chart corpus	Apache-2.0	`Ucas-HaoranWei/Vary-toy`	[98]
Fox	OCR-VLM	Focus-anywhere multi-page parser (Vary follow-up)	Doc + chart synthetic	Apache-2.0 (paper)	PENDING-PIN	[99]
TextMonkey	OCR-VLM	Monkey (Qwen-VL-derived) with shifted-window + token resampler	LAION + DocVQA + synthetic text	Apache-2.0	`echo840/Monkey` family	[100]
mPLUG-DocOwl1.5	OCR-VLM	mPLUG-Owl2 + Unified Structure Learning	DocStruct4M	Apache-2.0	`mPLUG/DocOwl1.5`	[101]
DocOwl2	OCR-VLM	mPLUG-DocOwl2 (high-res compression)	DocStruct + DocReason	Apache-2.0	`mPLUG/DocOwl2`	[102]
Mathpix	OCR-propriet-	Proprietary CNN+Trans-	Proprietary	Commercial / closed	API `v3`	[103]

Model	Family	Base architecture	Training data	License	Version pin	Citation
	ary (math)	former for STEM/equations				
DeepSeek-OCR / DeepSeek-OCR-2	OCR-VLM	DeepSeek-VL2 base + OCR head ("contexts optical compression")	Internal multimodal corpus	MIT (weights)	`deepseek-ai/DeepSeek-OCR`	[104]
GLM-OCR	OCR-VLM	GLM-4V derived OCR variant	ZhipuAI internal	Custom non-commercial-friendly	PENDING-PIN	[105]
FireRed-OCR	OCR-VLM	Xiaomi FireRedTeam OCR VLM	Internal (Xiaomi)	Apache-2.0 (per HF card)	`FireRedTeam/FireRedOCR` PENDING-PIN	[106]
Logics-Parsing-v2	Parse-intent VLM	Logics layout-aware parser (Qwen2.5-VL backbone reported)	Internal doc corpus	PENDING-LICENSE	v2 PENDING-PIN	[107]
Chandra (`datalab-to/chandra-ocr-2`)	OCR-VLM	Datalab Chandra-OCR-2 (PaIRS-leading per DOSSIER §2.4)	Datalab internal	Commercial / closed weights	HF `datalab-to/chandra-ocr-2` (DOSSIER pin)	[108]
DocIntentOCR-3B	Parse-intent VLM	3B doc-intent fine-tune (lineage PENDING)	PENDING	PENDING-LICENSE	PENDING-PIN	[10]
Qianfan-OCR	OCR-proprietary	Baidu Qianfan platform OCR endpoint	Internal Baidu	Commercial API	Qianfan API (PENDING-VERSION)	[109]
HyperAPI-PaddleOCR-OURS	OCR-pipeline	HyperAPI `parse` primitive over PaddleOCR-v4	Inherited from PP-OCRv4	Internal (Apache-2.0 upstream)	`hyperapi-sdk` PENDING-PIN	[10]
Qwen2-VL-7B / Qwen2-VL-72B	VLM	Qwen2 LLM + ViT encoder (Naive Dynamic Resolution)	Qwen2-VL multimodal mix	Apache-2.0 (7B) / Tongyi-Qianwen (72B)	`Qwen/Qwen2-VL-{7B, 72B}-Instruct`	[110]
Qwen2.5-VL-3B / -7B-Instruct	VLM	Qwen2.5 LLM + improved ViT (window attn)	Qwen2.5-VL mix; OCR-heavy stage	Apache-2.0 (3B/7B)	`Qwen/Qwen2.5-VL-{3B,7B}-Instruct`	[111]
Qwen3-VL-32B-Instruct / Qwen3-VL-235B	VLM	Qwen3 LLM + next-gen visual encoder	Qwen3-VL mix	Apache-2.0 (32B); larger sizes Tongyi-Qianwen	`Qwen/Qwen3-VL-32B-Instruct`, `-235B-A22B`	[71]

Model	Family	Base architecture	Training data	License	Version pin	Citation
Qwen3.6-A3B / Qwen3.6-35B-A3B	VLM (MoE)	Qwen3.6 35B-total / 3B-active MoE VLM	Qwen3.6 mix	Apache-2.0 (per card)	`Qwen/Qwen3.6-VL-A3B` PENDING-EXACT-PIN	[112]
Qwen3.5-2B-FT	VLM (fine-tune)	Hyperbots/in-house FT of Qwen2.5/3.x 2B variant	Internal	Internal	PENDING-PIN	[10]
InternVL2-8B / -26B / -76B	VLM	InternViT-6B + InternLM2 LLM (8B/20B/70B)	InternVL multimodal SFT mix	MIT (8B/26B), custom (76B)	`OpenGVLab/InternVL2-{8B, 26B,Llama3-76B}`	[72]
LLaVA-1.6-34B	VLM	Nous-Hermes-2-Yi-34B + CLIP-ViT-L	LLaVA-1.6 SFT mix	Apache-2.0 (weights), data CC-BY-NC	`liuhaotian/llava-v1.6-34b`	[113]
Llama-3.2-90B-Vision	VLM	Llama-3.1-70B-text + vision adapter, scaled to 90B	Meta multimodal SFT mix	Llama 3.2 Community License	`meta-llama/Llama-3.2-90B-Vision-Instruct`	[114]
GPT-4o	VLM (closed)	OpenAI native multimodal	Proprietary	Commercial API	`gpt-4o-2024-11-20` (DOSSIER-pinned PENDING-CONFIRM)	[115]
GPT-4o-mini	VLM (closed)	Smaller GPT-4o sibling	Proprietary	Commercial API	`gpt-4o-mini-2024-07-18`	[116]
GPT-4.1	VLM (closed)	Successor to GPT-4o	Proprietary	Commercial API (also via Azure OpenAI per [Hyperbots Azure OpenAI routing policy])	`gpt-4.1-2025-04-14`	[117]
GPT-5 / GPT-5.4 (thinking)	VLM (closed)	OpenAI GPT-5 family	Proprietary	Commercial API; default OpenAI route per [Hyperbots OpenAI-model preference policy]	`gpt-5.4-thinking` PENDING-CONFIRM	[118]
Gemini-1.5-Flash / Pro	VLM (closed)	Gemini 1.5 MoE multimodal	Proprietary	Commercial API	`gemini-1.5-{flash,pro}-002`	[119]
Gemini-2.5-Flash / Pro	VLM (closed)	Gemini 2.5 family	Proprietary	Commercial API		[120]

Model	Family	Base architecture	Training data	License	Version pin	Citation
					`gemini-2.5-{flash,pro}` PENDING-DATE	
Claude-Sonnet-4.6	VLM (closed)	Anthropic Claude Sonnet 4.6	Proprietary	Commercial API	`claude-sonnet-4-6` PENDING-CONFIRM	[121]
Mistral-Large-3-675B	LLM (closed weights)	Mistral Large 3, 675B (vision-enabled)	Proprietary	Mistral Research / Commercial	`mistral-large-3-2026` PENDING-CONFIRM	[122]

A.2. Verified-via-DOSSIER-only; deep-citation PENDING

The following entries appear in DOSSIER §2.1/§2.2 but were not re-verified against a canonical model card or paper at table-build time. Treat lineage as **unconfirmed** until a follow-up pass attaches a primary citation:

- **Qwen3.5-2B-FT**: internal fine-tune; base ambiguous (Qwen2.5-VL-2B vs Qwen3-VL micro).
- **GLM-OCR**: multiple ZhipuAI variants exist; need exact HF repo + commit.
- **Logics-Parsing-v2**: vendor (Alibaba/Logics) lineage and license PENDING.
- **DocIntentOCR-3B**: backbone unidentified.
- **Qianfan-OCR**: Baidu API endpoint; underlying model not separately published.
- **FireRed-OCR**: Xiaomi card lists Apache-2.0 but training-data disclosure is partial.
- **Chandra-OCR-2 (canonical description).** Vendor: Datalab (`datalab-to/chandra-ocr-2`), proprietary / commercial-closed weights distributed via Hugging Face under the DOSSIER pin; served in the framework as an OpenAI-compatible vLLM endpoint [39][108]. Architecture: an OCR-specialist VLM whose distinguishing property is **native bbox-grounded HTML emission**, every text block, table, and image is rendered as `<div data-bbox="x0 y0 x1 y1" data-label="...">` with a block-type label; per a 2026-05-26 smoke verification, Chandra is the first parse engine in scope to natively emit all four ParseBench primary metric surfaces (bbox_iou, layout_element_accuracy, reading_order_accuracy, GTRM) without prompt-engineering scaffolding [25][39][34]. **PaIRS table-split results (ParseBench n=10):** mean PaIRS 0.6472 (vs Qwen3.6-A3B 0.4422, Tesseract 0.3639), with documented high intra-model variance, per-doc range **0.2614–0.9996**, financial tables score up to 0.9996, whereas grid-heavy timetable-style layouts collapse to approximately 0.32 [10][25]. **Latency/throughput:** p50 approximately 8.5 s/page at 528 prompt + 821 completion tokens in the smoke configuration; this is the slowest of the OSS-local OCR-VLMs profiled [ofoxocr2026, parsexframework2026]. **Contamination posture:** no public training-data card; PaIRS table corpora cannot be ruled out of distribution, and the wide per-doc variance is consistent with contamination with a mixed in-/out-of-distribution effect [10]. **Known failure mode (F11-DICT-LITERAL):** under the uniform `max_tokens=2048` ceiling, Chandra (and Qwen) truncate on dense text-content / text-formatting splits at 8–10 of 10 documents (PENDING-MAXTOKEN, see §gates), producing 0.0 scores by truncation rather than capability. Lineage row: see Appendix A.1, line entry `Chandra (datalab-to/chandra-ocr-2)`.
- **Mistral-Large-3-675B**: parameter count and vision-capability claim sourced from DOSSIER §2.2 only.

A.3. Patterns

Three patterns are evident across the 50-row catalogue:

1. **OSS VLM convergence on Qwen2.5 / Qwen3 backbones.** PaddleOCR-VL (ERNIE-based) and DeepSeek-OCR (DeepSeek-VL2) are the main exceptions; nearly every other open OCR-VLM since late 2024 (GOT-OCR-2.0, MinerU2.5-Pro, Logics-Parsing-v2, the Qwen3.6-A3B family itself) builds on Qwen2.5/3 LLM weights with a custom vision encoder or OCR head [111][71][97][32][104].

2. **Frontier closed models still dominate parse-intent.** On the inhouse `parse-intent-inhouse-v1` rubric (DOSSIER §2.2), Claude-Sonnet-4.6 / GPT-4.1 / Gemini-2.5-Pro sit in the top tier; the open Qwen3-VL-32B/235B family is the closest open challenger. Exact F1 values are omitted here because the rubric is Hyperbots-internal and not externalized [10].

3. **License heterogeneity constitutes a substantial adoption constraint.** Apache-2.0 dominates the small-model OSS column, but key pipeline tools (Marker/Surya GPL-3.0 dual; MinerU AGPL-3.0; Nougat CC-BY-NC) and the proprietary Tongyi-Qianwen / Llama-3.2-Community / Mistral-Research tiers each impose distinct deployment restrictions. A consolidated leaderboard cell that mixes these without a license column risks misrepresenting deployability.

Contamination notes

Several base models are documented (or strongly suspected) to have ingested datasets that are simultaneously used as evaluation corpora elsewhere in the field. These should be flagged whenever a Parse-x cell uses one of those corpora as the gold set:

- **PubTabNet / PubLayNet contamination risk.** Nougat is trained on arXiv PDFs whose tables overlap PubTabNet's IBM source distribution; GOT-OCR-2.0 and DocOwl1.5/DocOwl2 explicitly list "publicly available document understanding datasets" that the model cards do not enumerate exhaustively; PubTabNet / PubLayNet scores for these models should therefore be regarded as **potentially contaminated** [96][97][101].
- **OmniDocBench / OCRBench leakage risk.** MinerU 2.5, PaddleOCR-VL-1.5, Qwen2.5-VL, and InternVL2 all cite "OCR-heavy multimodal pretraining mixes" that, per the model cards, were assembled before OmniDocBench / OCRBench held-out splits were finalized; deduplication is not independently audited [95][32][111][72].
- **`results.zip` self-loop.** Per the benchmark-factory guardrail (CLAUDE.md §21, cited in DOSSIER §2.2), HyperAPI deployed output (`results.zip`) is **never** ground truth; any in-house fine-tune (Qwen3.5-2B-FT, HyperAPI-PaddleOCR-OURS) that was trained against HyperAPI-produced labels must be evaluated only on independently-labeled corpora [10].
- **Chandra-OCR-2 on PaIRS.** No public training-data card; we cannot rule out PaIRS-relevant table corpora being in-distribution. DOSSIER §2.4 already notes high per-doc variance (0.26–0.9996), which is consistent with a mixed in- and out-of-distribution effect [10].

Action item for the next dossier iteration: convert every "PENDING-PIN" and "PENDING-LICENSE" entry above into a verified commit hash and SPDX identifier, and add a contamination-audit column tied to the gold corpus of each Parse-x dimension.

Appendix B. Datasets & Contamination Posture

This appendix inventories every public and internal corpus that a Parse-x leaderboard cell could draw on, records what is known about its provenance and possible model-training overlap, and pins each to one or more of the framework's seven measurement dimensions (D-OCR, D-PARSE-INTENT, D-LAYOUT, D-BBOX, D-TABLE, D-DOWNSTREAM, D-COST-LATENCY) and two stratification overlays (D-DOC-QUALITY, D-LANG) [25]. The appendix then states the standing circularity rule on `results.zip` and closes with the per-doc-type coverage-gap matrix.

Contamination posture follows a three-band scheme. **Known-train:** a primary source documents that the corpus appears in a surveyed model's training data. **Plausible-train:** the corpus has been on the public web long enough or in a popular HF mirror such that web-scale VLMs likely saw it, but no training-data card confirms it. **Held-out / internal:** corpus is private, paywalled, or post-dates the cutoff of all surveyed models. Per-claim citations follow.

B.1. Public datasets

Dataset	n_docs / n_rules	License	Source	Contamination known?	Framework dim
ParseBench (5 splits: chart, layout, table, text_content, text_formatting)	~169k rules across 5 splits; n=10/split currently on disk	Apache-2.0	upstream `hyprbots/parsebench`, commit a0b10dd7 [49]	Plausible-train for any model post-dating its 2024 release; rule files indexed by HF	D-PARSE-INTENT, D-TABLE, D-LAYOUT, D-OCR (text_content)
SAVIOR-Bench v1	509 expert-annotated docs; corpus mix UCSF 46.95% / synthetic 17.88% / misc. public financial 17.49% / Inv3D 9.04% / FUNSD 8.64%; IAA 0.761 (fuzzy-string) over 187 double-annotated pairs; attested 31-system OCR edit-distance leaderboard; documented R1-R10 failure taxonomy; per-doc score vectors for the attested leaderboard not on disk (open gate R2 / BLK-18)	research/ non-commercial (per paper)	SAVIOR-Bench §2.1 [1][123]	Held-out at release; constituent sub-corpora (FUNSD 8.64%, Inv3D 9.04%) are long-public and almost certainly in VLM pre-training	D-OCR, D-PARSE-INTENT, D-LAYOUT (via Inv3D share), D-BBOX (Inv3D upstream boxes), D-DOC-QUALITY (low-quality class label), D-LANG (multilingual residual cluster)
PubTabNet	568k table images (PubMed Central)	CDLA-Permissive-2.0	IBM Research [45]	Known-train for nearly every table-structure model (Table-Transformer, GTRM-family, PaddleOCR-VL)	D-TABLE
FinTabNet	~113k financial tables (S&P annual reports)	CDLA-Permissive-2.0	IBM Research [46]	Known-train for table-structure VLMs trained on the IBM mix	D-TABLE

Dataset	n_docs / n_rules	License	Source	Contamination known?	Framework dim
DocLayNet	80,863 manually labelled page images, 6 doc classes	CDLA-Permissive-1.0	IBM/DS4SD [27]	Known-train for layout detectors (LayoutLMv3 fine-tunes, DocLayout-YOLO)	D-LAYOUT, D-BBOX
PubLayNet	358k pages from PubMed Central	CDLA-Permissive-1.0	IBM Research [26]	Known-train for most layout backbones	D-LAYOUT, D-BBOX
DocVQA	50k questions over 12,767 industry document images; answer-string accuracy; recent direct-VLM systems (Agentic Document Extraction) reach 99.16, outperforming OCR pipelines by 3.2 pp GT-in-Pred [78][79]; scores answers, not field-set micro-F1, and varies extractor and parser jointly	non-commercial research	Mathew et al. [78][124]	Known-train for InternVL, Qwen-VL, GPT-4o (publicly stated); also appears bundled in DUE [19]	D-DOWNSTREAM (extractive QA), D-PARSE-INTENT
OCRBench v2	10,000 instances / 23 tasks / 31 scenarios	research	Fu et al. [3]	Plausible-train; many constituent sub-tasks are public	D-OCR, D-PARSE-INTENT
Omni-DocBench	981 PDF pages, 9 doc types, 4 layout types	Apache-2.0	OpenDataLab/ MinerU team [4]	Known-train for MinerU-2.5 (acknowledged in MinerU technical report)	D-LAYOUT, D-TABLE, D-OCR
olmOCR-Bench / olmOCR-mix	~7k mixed-source pages	Apache-2.0	AllenAI [126]	Held-out at release; Qwen-2-VL-7B-FT model trained on companion `olmOCR-mix-0225`	D-OCR
ChartQA	9,608 human + 23,111 machine QA over 21k charts	GPL-3.0	Masry et al. [127]	Known-train for chart-capable VLMs	D-PARSE-INTENT (chart), D-DOWNSTREAM
FUNSD	199 noisy scanned forms (149 train / 50 test); 9,707 semantic entities in 4 classes (header/	research only	Jaume et al. [9] [77]	Known-train, widely used since 2019; appears as 8.64%	D-LAYOUT, D-BBOX, D-PARSE-INTENT (form KIE)

Dataset	n_docs / n_rules	License	Source	Contamination known?	Framework dim
	question/answer/other) over 31,485 word-level boxes with entity links; reference task Semantic Entity Recognition (entity-level F1), LayoutLMv3-large ~92, FormNetV2 86.35 [12] [22]; single-page English forms only			of SAVIOR-Bench [123]	
CORD	1,000 Indonesian receipts (800 / 100 / 100); 30 entity types in 4 groups (Menu / Void menu / Subtotal / Total); field-level F1 + tree-edit-distance accuracy; LayoutLMv3-large 97.46 F1, Donut 91.6 [12][14]; receipt-only, short documents, single language	CC-BY-4.0	Park et al. [13] [74]	Known-train for Donut and most receipt models	D-OCR, D-PARSE-INTENT
SROIE (ICDAR 2019, Task 3)	973 scanned restaurant receipts (626 train / 347 test); fixed 4-field schema (company / date / address / total); field-level F1, LayoutLM-large 95.2 [76]; the de facto entry-level KIE benchmark, with a small fixed schema far below the 23-system, free-schema inhouse-v1 panel	research only	Huang et al. [15] [75]	Known-train; in many VLM doc mixtures	D-OCR, D-BBOX, D-PARSE-INTENT (fixed-schema KIE)

Notes on apples-to-apples comparability: ParseBench n on disk is 10 docs per split [10]; the upstream `n_rules` figure refers to rule cardinality rather than document count, and the two should not be conflated when reporting confidence intervals. SAVIOR-Bench attested per-doc score vectors are not on disk, blocking interval claims for the WordRecall reconciliation gate (BLK-18) [10].

B.2. Internal Hyperbots datasets

Treated as **internal corpus** unless a public attestation exists. Per-row numerics stay inside the factory and are not externalized without consent (per team guardrails).

- **dataset_95 (`dataset_95_granular`)**: 95 reviewed invoices/statements; gold has `summary` + `line_items` + `number_of_line_items`; `_status: reviewed`, `_source: sub-`

`agent_validated`. Hold-out status PENDING [10]. Primary use: D-DOWNSTREAM (extract FieldRecall), D-COST-LATENCY (300s timeout characterization).

- **financepss-205**: 205 finance docs; 49 attested rows; per-doc score vectors absent (BLK-3) [10]. Use: D-DOWNSTREAM.
- **Edit-distance OCR corpus** (`DC-000__ocr-edit-distance-v1`): 31 measured cells; per-corpus n not in attested source. Use: D-OCR.
- **Parse-intent in-house rubric** (`DC-000__parse-intent-inhouse-v1`), 19 cells; rubric provenance vs. `results.zip` PENDING (Q-DOSS-2), must be cleared before D-PARSE-INTENT publication [10].
- **SOULSHINE BoL**, 44 docs (soulshine1: 22, soulshine2: 22; soulshine2 byte-identical to bol-grn-test inputs); no gold (uses field-axis proxy). D-DOWNSTREAM-adjacent only.
- **tough/bank_statement**, 44 docs; median 12pp, max 51pp (one borderline for the 300s timeout). D-DOWNSTREAM, D-COST-LATENCY.
- **tough/cheque**, 21 docs, PNG-only. D-OCR.
- **tough/credit_note**, 34 docs, PNG-only. D-DOWNSTREAM.
- **tough/remittance**, 75 docs (37 PDF + 38 PNG); closest of any tough split to the n=100 threshold (+25 needed) [128]. D-DOWNSTREAM.

All internal corpora share one contamination caveat: because Hyperbots' own product is a parse/extract pipeline, gold labels in any internal corpus bootstrapped from product output risk circularity for HyperAPI cells. Each internal corpus must carry an explicit `gold_kind` flag, see B.3.

B.3. Circularity and the `results.zip` rule

The framework defines a `gold_kind` enum [25] whose `"results-zip-derived-NEVER-GOLD"` value is the standing prohibition: any artifact derived from `results.zip`, the HyperAPI deployed-IDP output bundle, is forbidden as ground truth. The rule is restated in three places: the factory's `CLAUDE.md` ("`results.zip` is HyperAPI deployed output; NEVER used as ground truth (circular)") [129]; the framework header ("`results.zip` IS HyperAPI's deployed-IDP output and is circular for Hyperbots and biased toward Hyperbots' field schema for competitors") [25]; and the dossier's parse-intent §2.2 (forcing a circular-GT check on the inhouse-v1 rubric) [10].

The implication is consequential: using `results.zip` as gold would (i) guarantee HyperAPI a near-perfect score against itself, inflating any "best for all finance" claim under H0, and (ii) penalize competitors whose schemas legitimately diverge from Hyperbots'. The sanctioned uses are limited to schema-anchor / drift detection [25]. Two implications follow for this review. First, D-PARSE-INTENT cells using the in-house rubric carry a publishability asterisk until Q-DOSS-2 closes. Second, the framework's `dimension D-DOWNSTREAM` baseline of "Qwen 3.6 on raw image" is preferred over "Chandra-as-baseline" precisely to avoid a second circularity, since Chandra appears as a parse engine in the panel [25].

B.4. Per-doc-type coverage gap matrix

From `DOC-TYPE-GAPS.json` (2026-05-26) [128], counting Tier-1+2+3+4 doc types against the n ≥ 100 minimum: **5 of 6 Tier-1 doc types currently have ZERO factory-measured cells** (T-02 credit note, T-03 PO, T-04 packing slip, T-05 BoL, T-06 remittance; only T-01 invoice has measured cells, with 5 vendors on n=90–95). **Tier-2 has zero measured cells across all four types** (T-07 bank statement, T-08 cheque, T-09 pay stub, T-10 wire/ACH). **Tier-3 has zero measured cells**, and T-11 (10-Q/10-K) is structurally excluded from HyperAPI IDP's current doc-type set [doctypegaps2026, BLK-7]. **Tier-4** (T-14 insurance, T-15 loan) has neither corpus nor measured cells. Of the 16 framework doc types, **14 currently have zero leaderboard cells** and **2** (T-01 invoice and T-16 FUNSD-style forms via SAVIOR aggregation) have any measured presence. Summary for Parse-x: the leaderboard today is a single-doc-type (invoice) result with an OCR-panel aggregate over SAVIOR; cross-doc-type claims are unsupported until D2/D3 phases run.

Appendix C. Vendor & API Pin Table

This appendix pins the closed-source / hosted services in the Parse-x panel as of 2026-06-16. Where a vendor has shipped multiple GA snapshots in the past quarter, the table lists the snapshot pinned by the harness; `PENDING-CONFIRMATION` flags entries where the public documentation is ambiguous or where a sandbox call to confirm the snapshot string has not yet been executed.

Vendor	Service	Current GA model(s)	API version pin	Auth	Pricing	Region	Citation
Microsoft	Azure Document Intelligence (fmr. Form Recognizer)	`prebuilt-layout`, `prebuilt-read`, `prebuilt-document`, `prebuilt-invoice`, custom-neural v4	REST `2024-11-30` (GA, v4.0)	Key (subscription) or Entra ID	from $1.50 / 1k pages (Read S0); Layout S0 $10 / 1k pages	30+ Azure regions; not all prebuilts in every region	[130], [131]
Google	Gemini API (AI Studio / Vertex)	`gemini-2.5-pro`, `gemini-2.5-flash`, `gemini-2.5-flash-lite`; `gemini-3.1-pro` and `gemini-3.5-flash` rolling out	`gemini-2.5-pro-preview-06-05` for Pro panel; 2.5-flash GA snapshot, `PENDING-CONFIRMATION` for 2.6/3.x in our harness	API key (AI Studio) / OAuth + ADC (Vertex)	2.5 Pro $1.25 / $10 per 1M tok (≤200k ctx); 2.5 Flash $0.30 / $2.50 per 1M tok	global, us-central1, europe-west4, asia-south-east1	[132], [133]
Anthropic	Claude API	`claude-opus-4-8` (flagship, rel. 2026-05-28), `claude-opus-4-7`, `claude-sonnet-4-6`, `claude-haiku-4-5`	Pin: `claude-sonnet-4-6-20260415`, `claude-opus-4-8-20260528` (PENDING-CONFIRMATION of exact dated snapshot for 4.8); 4.5 family deprecated path; `anthropic-version: 2023-06-01` header	API key	Sonnet 4.6 $3 / $15 per 1M; Opus 4.7/4.8 $5 / $25 per 1M; Haiku 4.5 $1 / $5; batch -50%, cache -90%	us-east, eu, AWS Bedrock + GCP Vertex mirrors	[134], [135]
OpenAI (via Azure)	Azure OpenAI Service	`gpt-5` (2025-08-07), `gpt-5.1` /	Data-plane: `api-version=2025-04-01-preview` for Re-	Entra ID (pre-	Mirrors OpenAI list price + Azure	eastus, eastus2, sweden-	[136], [137], [138]

Vendor	Service	Current GA model(s)	API version pin	Auth	Pricing	Region	Citation
		`gpt-5.1-chat` (2025-11-13), `gpt-5.2-chat` (2026-02-10), `gpt-5.4` thinking/pro (2026-03-05), `gpt-5.4-mini/-nano` (2026-03-17), `gpt-4.1`, `gpt-4o`, `gpt-4o-mini`, `o3` (2025-04-16), `o3-mini`	sponses API; deployment-pinned snapshots above. Older GPT-5/o3 snapshots deprecate 2026-12-11	ferred) or key	markup; GPT-5.4 thinking ~ $5 / $20 per 1M tok (PENDING-CONFIRMATION on Azure-side rate card)	central, japaneast, australiaeast (model-dependent)	
OpenAI direct	NOT USED in our harness	,	,	,	,	,	see C.1
Mistral	La Plateforme	`mistral-large-3` (`mistral-large-2512`), `mistral-medium-3`, `mistral-small-3.2`	`mistral-large-2512`	Key	Large 3: $2 / $6 per 1M tok (one source); $0.50 / $1.50 for `large-2512` variant, discrepancy, `PENDING-CONFIRMATION` against vendor card	EU (Paris), US via AWS Bedrock	[139], [140]
Mathpix	Convert API (v3)	`mathpix-ocr` (text+math); `pdf` endpoint	REST `v3`	App-ID + App-Key headers	$0.004–$0.01 / page (Convert PDF); $0.005 / image equation OCR, `PENDING-CONFIRMATION` on 2026 rate card	global (US-hosted)	[141]
LlamaIndex	LlamaParse (LlamaCloud)	modes: `fast`, `balanced`, `premium`, `agentic`	LlamaCloud API `2024-12` GA; `agentic` mode GA Q1 2026	API key	Credit system: 1,000 credits = $1; `fast` 1 cr/pg, `premium` 15 cr/pg, `agentic` 45	US (AWS us-east-1), EU	[142], [143]

Vendor	Service	Current GA model(s)	API version pin	Auth	Pricing	Region	Citation
					cr/pg; 10k free credits/mo		
Reducto	Reducto Parse / Extract	`parse`, `extract`, `split`, `edit` agentic endpoints	`2025-11` GA; pin `model=parse-v2`	API key (bearer)	Credit system: 1,000 credits = $1; recommended preset ~$0.01–$0.03/pg	US (SOC2); on-prem option	[144]
Unstructured.io	Serverless API / Platform	`unstructured-api` v0.x; partition strategies `hi_res`, `auto`, `fast`, `ocr_only`, `vlm`	API `2025-09`; pin `strategy=hi_res, hi_res_model_name=yolox`	API key	Serverless: $1 / 1k pages (`fast`), $10 / 1k pages (`hi_res`); free OSS	US, EU; OSS = anywhere	[145], [146]
IBM	Docling	OSS library (`docling` PyPI), Docling-Serve container, watsonx.ai-hosted	Pin: `docling==2.x` (commit SHA in harness lockfile)	none (local) / IAM (watsonx)	OSS = free; watsonx.ai = compute-billed	self-host; watsonx multi-region	[147], [146]

C.1. Azure OpenAI routing rule

All OpenAI model calls in Parse-x (`gpt-4o`, `gpt-4o-mini`, `gpt-4.1`, `gpt-5`, `gpt-5.1`, `gpt-5.2`, `gpt-5.4` family, `o3`, `o3-mini`) are routed through **Azure OpenAI Service** rather than the direct `api.openai.com` endpoint. This is a standing Hyperbots project preference, recorded in user memory `azure-openai-preferred` and reaffirmed in `openai-default-gpt-5-4-thinking` (current default = GPT-5.4 thinking mode) [148]. Operational reasons: (i) Entra-ID auth and tenant-scoped data residency, (ii) committed-throughput PTUs for reproducible latency cells in D-COST-LATENCY, (iii) the Azure data-processing addendum the Hyperbots tenant has signed. In practice, model snapshot strings follow Azure's deployment-name convention, the API path is `/openai/deployments/{deployment}/responses?api-version=…`, and snapshot availability typically lags OpenAI direct by one to six weeks [136]. Benchmark cells that name an OpenAI model implicitly mean "this snapshot, served via Azure"; when an OpenAI snapshot is not yet available on Azure, the cell is marked `BLOCKED-AZURE-LAG` rather than falling back to direct OpenAI.

C.2. Known boundaries & caveats

- **Azure DI** prebuilt-read covers Latin and CJK printed text, but handwritten Chinese, Japanese, and Korean fall back to print-trained models with degraded recall; handwritten Arabic is not supported in `prebuilt-read` GA [130]. The `prebuilt-layout` table-cell linking does not emit row/column spans for merged headers in all layouts; this is relevant to D-TABLE.
- **Gemini 2.5 / 3.x** vision input is billed as tokens (per Google's tokenizer for images at ~258 tok/tile), so D-COST-LATENCY normalization must convert pages to tokens before cost comparisons against per-

page services. The `2.6` / `3.x` line had not stabilized at the cutoff date; the pin is therefore marked `PENDING-CONFIRMATION` [132].

- **Claude Opus/Sonnet** 1M-token context is flat-rate on 4.6/4.7/4.8 (no surcharge), which changes the cost calculus for whole-PDF prompting vs page-tiling [134]. Sonnet/Opus 4.5 snapshots remain callable but are on a deprecation path and are not pinned in this review.
- **GPT-5.4 thinking / pro** exposes reasoning-token billing separate from completion tokens; D-COST-LATENCY must report both. Snapshot deprecation for older GPT-5/o3 takes effect on 2026-12-11; pins must be refreshed before that date [137].
- **Mistral Large 3** is multimodal (MoE 41B active / 675B total) but vision-OCR head is newer than the text head; expect higher variance on D-OCR than D-PARSE-INTENT [139].
- **Mathpix** remains among the strongest systems surveyed for math-equation OCR but does not return layout bounding boxes at the same granularity as Azure DI, which affects D-BBOX scoring.
- **LlamaParse / Reducto / Unstructured** use credit-based or per-strategy pricing; to enable apples-to-apples comparison in D-COST-LATENCY, costs are normalized to "USD per 1,000 pages on the `hi_res` / `premium` / `parse-v2` preset", with the preset disclosed in every cell.

2.8 Industrial IDP landscape (vendor and analyst literature)

Running parallel to the academic document-parsing literature surveyed above is a substantial industry literature on Intelligent Document Processing (IDP) that this review touches only where it overlaps the seven measurement dimensions. The two literatures share an architectural pattern (a parse stage that lifts pixels to a structured intermediate, followed by an extract stage that maps the intermediate to business fields), but differ sharply on evaluation discipline.

The vendor pool typically named by analyst firms covers UiPath Document Understanding [172], Extend AI Parse 2.0 [188], Ocrolus [198], Nanonets [199], Sensible [200], Adobe Acrobat AI Assistant[202] (a layout-first parsing API trained on 1M+ real-world pages, with its companion RealDoc-Bench [189] benchmark of 1,500 real-world documents from healthcare, financial services, logistics and real estate that measures both layout accuracy and downstream agent performance), Hyperscience Hypercell [175], Rossum (now Coupa) [176], ABBYY (FlexiCapture, Vantage), Microsoft 365 document-processing services (the rebrand of Syntex) [177], Google Document AI [174], AWS Textract [173], Indico, Kofax/Tungsten, and Automation Anywhere Document Automation. A separate sub-class of invoice and receipt specialists, often grouped together by analyst firms, comprises Mindee [192], Veryfi [193], Affinda [194], Klippa [195], and the document-extraction API specialists Ocrolus [198], Nanonets [199] and Sensible [200]; these vendors publish their own field-level accuracy and per-page latency numbers but typically without the per-document score vectors or fine-tune-vs-zero-shot disclosure that the academic literature now expects. Analyst-firm taxonomies (Forrester Wave IDP, Gartner Magic Quadrant for IDP, Everest Group PEAK Matrix IDP, IDC MarketScape IDP) usually segment the field along three axes: (i) templated rules-and-zonal-OCR engines (the legacy ABBYY FlexiCapture lineage), (ii) ML-driven engines that train per-customer extraction models (Hyperscience, Rossum, Indico), and (iii) hybrid engines that bolt a pretrained transformer or, more recently, a vision-language model on top of a template fallback (UiPath DU, Microsoft, the hyperscaler services). The most recent vendor messaging (e.g., Hyperscience ORCA, UiPath generative extraction) collapses parse and extract into a single VLM call, which mirrors the end-to-end direction in the academic literature (Nougat, olmOCR, dots.ocr) but is marketed as a no-train default rather than as a fine-tuning recipe.

Three substantive deltas separate this industrial literature from the academic prior art covered above. First, evaluation discipline is closed: vendor white papers report customer-pilot headline accuracy ('99.5%

accuracy and 98% automation' is the canonical Hyperscience claim [175]) without per-document score vectors, bootstrap CI95, or fine-tune-vs-zero-shot disclosure, and analyst-firm scoring is composite-vendor capability rather than per-document benchmark. Second, the document mix is enterprise-AP-centric (invoices, POs, remittances, claims) rather than the academic mix of arXiv PDFs, layout corpora (DocLayNet, PubLayNet), and table corpora (PubTabNet, FinTabNet), so cross-comparison with academic numbers is not apples-to-apples. Third, cost reporting is opaque: pricing is per-page or per-document, sometimes credit-bundled, and rarely normalized to '$/correct call' in the D-COST-LATENCY sense. This review therefore treats the industry literature as taxonomic context and a source of vendor pins (Appendix C), not as a substitute for the per-document, per-stratum benchmark cells the framework requires.

- **Docling** is OSS; a commit SHA is pinned in the harness lockfile so that a `docling` upgrade does not silently change scores between leaderboard refreshes.
- Vendors' published prices are list prices; enterprise / committed-use discounts that may apply to Hyperbots' actual spend are NOT reflected here and MUST NOT be externalized.

2.11 First-pass measured baseline

Preliminary status. The numbers in this subsection are first-pass smoke measurements at n=10 per split on the held-out ParseBench corpus. They are reported here to anchor the prior-art survey in actually-observed deltas, not as a leaderboard. Three caveats bind every cell below: (i) n=10 implies a 2000-resample bootstrap CI95 half-width of approximately ±0.15 on a [0,1] metric (see §4.3), so most engine-to-engine deltas reported here are *not* statistically separable; (ii) PaIRS uses the private `hyprbots/vlm_ocr@1fbbc334` implementation and is therefore *internal-only*, external readers cannot reproduce the PaIRS column from this review alone (see §4.2); (iii) the F-B suite was executed under a uniform `max_tokens=2048` ceiling per the BLK-15 pin, which silently truncates several cells and converts a capability question into an output-length question (see §4.5). Decoding was pinned across all engines for both runs: `T=0.1`, `top_p=0.9`, `max_tokens=2048`, `seed=20260516`. The internal DH-001 / `dataset_95_granular` extract numbers are *not* included in this subsection and remain internal pending the hold-out determination noted in Appendix B.

Table 2.11.a, PaIRS panel, ParseBench table split (n=10, measured 2026-06-09; z-Euclidean, higher is better).

Engine	PaIRS	Per-document range	Notes
Chandra (`datalab-to/chandra-ocr-2`)	**0.6472**	0.2614 – 0.9996	Lead engine on D-LAYOUT reading-order surrogate; per-doc variance is the dominant story (range ≈ 0.74).
Qwen3.6-A3B	0.4422	,	Second; 0.2050 below Chandra (within the n=10 CI95 envelope).
Tesseract	0.3639	,	Plain-OCR primitive; structurally disadvantaged on spatial-relationship scoring.
HyperAPI parse (paddle)	0.2907	,	Lowest; structurally expected, the primitive emits flat markdown with no spatial geometry.

Per-dimension lead engine: **Chandra leads D-LAYOUT (PaIRS)** in this first pass. The Chandra ↔ Qwen3.6-A3B gap (0.2050) is comparable to the statistical floor and should not be reported as a settled ranking; the Chandra ↔ HyperAPI-parse gap (0.3565) reflects a primitive-type difference rather than a within-class engine comparison.

Table 2.11.b, F-B suite valid cells, ParseBench diverse-doc splits (n=10 unique PDFs per split, measured 2026-06-10). Only cells that are non-trivial *and* not invalidated by truncation or structural mismatch are shown; the rest of the panel is documented in DOSSIER §2.5 and is suppressed here to avoid implying signal where none exists.

Split / metric	Chandra	Qwen36-a3b	Tesser-act	Hyper-API parse	Lead	Apples-to-oranges flag
PB-002 layout / ElementPassRate	**0.2000**	0.0000	0.0000	,	Chandra	Qwen `t=10/10` length-truncation at `max_tokens=2048`, cell is not a capability measurement.
PB-004 text_content / ContentFaithfulness (`missing_sentence_percent`)	0.4653	**0.5584**	0.3651	0.2073	Qwen36-a3b	Qwen and Chandra both touch the 2048-token ceiling on dense pages; absolute scores understate the VLM engines relative to OCR primitives.
PB-005 text_formatting / SemanticFormatting	0.6000	**0.9000**	0.0000	0.0000	Qwen36-a3b	Tesseract and HyperAPI-parse 0.0 are structural (plain-OCR primitives do not emit HTML/ markdown formatting tags), not engine failure.

Per-dimension lead engines from valid F-B cells: **Chandra on layout/ElementPassRate; Qwen3.6-A3B on text_content/ContentFaithfulness and text_formatting/SemanticFormatting; Tesseract second on text_content (a result that should not be over-interpreted, at n=10 with CI95 half-width ≈ ±0.15 the Tesseract ↔ Qwen gap is roughly one floor-width).** The chart split (PB-001), table split (PB-003 / GTRM), and the remaining text-content cells were either all-zero by structural mismatch or all-zero by `max_tokens=2048` truncation across at least 7/10 documents; per the BLK-15 escalation those cells are categorically excluded from this subsection rather than reported as 0.0.

Apples-to-oranges flag (carried into §4.5). Two systematic distortions are visible in Table 2.11.b. First, the uniform `max_tokens=2048` ceiling is *not* a fair pin across engine classes: Tesseract and HyperAPI-parse complete every document under the budget (they emit short flat text), while Chandra and Qwen3.6-A3B routinely exceed it on dense financial pages and are scored on a truncated suffix. Second, the OCR primitives score 0.0 on every layout/table/formatting metric because they do not *attempt* the structured output the metric checks for; this is not a capability gap inside the OCR-primitive class but a category mismatch with the metric contract. Both effects inflate the apparent OCR-primitive performance on text_content and depress the apparent VLM performance on layout/formatting; §4.5 treats this as the central cross-engine fairness caveat and the principal motivation for raising the `max_tokens` ceiling before any headline cell is published.

Forward references. The statistical-floor implication (n=10 → CI95 half-width ≈ ±0.15) is developed in **§4.3**, which argues for widening n before any headline cell is externalized. The truncation and metric-contract distortions called out above are taken up in **§4.5** as the canonical apples-to-oranges fairness note for the framework. Until those two gates close, the numbers in this subsection should be read as a *baseline of measurement plumbing* (the cells exist, the decoding is pinned, the corpus is on disk) rather than as a comparative claim about the engines themselves.

3. Discussion and Open Gates

The discussion below is deliberately brief; substantive limitations, reproducibility caveats, and the statistical-power floor are promoted to Section 4 (Limitations), in accordance with NeurIPS-D&B convention.

4. Limitations

Three patterns emerge across the seven dimensions and two overlays. First, **per-document score vectors are absent everywhere except SAVIOR-Bench's internal pipeline**, OCRBench v2, OmniDocBench, olmOCR-Bench, PubTabNet, and DocLayNet all publish single aggregate numbers per model; consequently, any rigorous bootstrap CI95 reporting built on top of them requires either re-running the evaluation (with the attendant compute and decoding-pin disclosure burden) or explicit attestation. Second, **fine-tune-vs-zero-shot disclosure is inconsistent in the published literature**; this is a known SAVIOR-Bench complaint (R6) and a recurring source of misleading model-vs-model comparisons (e.g., a 2B fine-tuned model outperforming a 75B zero-shot model is structurally expected and should not be reported as evidence of superior model capability). Third, **parse-to-downstream lift (D-DOWNSTREAM) is under-used in the parse-benchmark literature**: prior art treats OCR and parse as an intrinsic-metric end-state rather than as a stage feeding extract, RAG, and classify pipelines, and the paired-bootstrap testing infrastructure imported from NLP [73] and ASR (formalised for NLP by [164]), together with the RAG-evaluation lineage [166], has not been carried over.

The largest open gates blocking first-pass cell publication, in order of severity, are: (i) BLK-18, the SAVIOR-Bench 509-doc out7 corpus is not on disk, blocking WordRecall reconciliation against the attested 31-cell leaderboard; (ii) PENDING-MAXTOKEN, the uniform max_tokens=2048 ceiling truncates Chandra and Qwen on dense text-content / text-formatting splits at 8-10 out of 10 documents, producing 0.0 scores by truncation rather than by model capability; (iii) the model-lineage table (Appendix A) carries several PENDING-PIN entries that block the per-cell `pins.adapter_version` field from being populated; (iv) several Tier-1 document types in DOC-TYPE-GAPS still have zero measured cells across every framework dimension, a gap that no public benchmark fills.

The framework's measurement discipline does not, by itself, resolve any of these gates; rather, it renders them visible. This visibility, combined with a curated and rerunnable cell registry, informs the industrial release discipline that follows.

4.1 Scope limitations

This review is deliberately scoped to the seven measurement dimensions named in §2 and treats four content surfaces as out of scope:

- **Handwritten / cursive recognition** (formerly out-of-scope; brief coverage now in section 2.8). IAM database [178], READ / ICFHR competitions, and RIMES define the benchmark lineage; handwritten recognition exhibits distinct failure modes and is not directly addressed by the printed-text engines surveyed here.
- **Mathematical formula extraction** (Im2Latex, CROHME competitions, MathPix-style image-to-LaTeX). Specialist sub-field with its own gold corpora and structured-output conventions.
- **Document classification** (RVL-CDIP and similar). Treated upstream of parse rather than inside it.

2.9 Adjacent surfaces (math formula extraction + document classification)

Two content surfaces are designated out-of-scope in §3.1 but warrant a brief survey, since adjacent vendors routinely conflate them with general document parsing. This subsection summarizes prior art for each, names canonical benchmarks and modern engines, and states the explicit delta from the Parse-x primary parsing and extraction surface.

(a) Mathematical formula extraction. The image-to-markup line was opened by Im2Latex / Im2Markup, which framed formula recognition as a seq2seq problem with a coarse-to-fine visual attention encoder and decoded LaTeX tokens directly [183]. The CROHME competition lineage, hosted at ICDAR since 2011 and most recently summarized in the 2019 CROHME + TFD competition report, supplies the canonical stroke-based and image-based handwritten-formula corpora (online InkML traces, offline rendered images) together with task-specific structure-recognition metrics (expression-rate, symbol-rate) [184]. Modern open implementations such as pix2tex / LaTeX-OCR pair a Vision Transformer encoder with a Transformer decoder and report BLEU 0.88 / normalized edit distance 0.10 on typeset formula crops, with closed equivalents at MathPix and inside several VLM stacks [185]. Delta from Parse-x: formula extraction assumes the formula region has already been detected and cropped, its gold output is LaTeX (a structured-output convention that the Parse-x text-content metric is not designed to score), and its failure modes (subscript/superscript scope, fence matching, accent placement) do not occur in the Parse-x text-content or text-formatting splits. Parse-x cells therefore exclude formula-only documents, although mixed financial-table pages containing inline math are scored against the rendered LaTeX glyph string, not against parsed LaTeX.

(b) Document classification. The dominant benchmarks here are RVL-CDIP (400,000 images, 16 classes, drawn from the IIT-CDIP collection and released by Harley et al.) and the smaller Tobacco-3482 (3,482 documents across 10 classes, Kumar et al.), both sourced from the Truth Tobacco Industry Documents archive of legal-discovery scans [186] [187]. Modern engines are typically vision-only CNNs or VLM zero-shot classifiers, with transfer-learning recipes from RVL-CDIP to Tobacco-3482 as the de-facto leaderboard. Delta from Parse-x: classification asks "what kind of document is this" (one label per image), whereas Parse-x measures within-document parse and extract quality (text, layout, bbox, table, downstream field recall) conditional on the document type being known. The two pipelines are sequential, not competing: a production stack typically classifies first, then routes to a parse profile. Parse-x's sixteen-document-type taxonomy is consumed from upstream classifiers; building or evaluating those classifiers is left to the RVL-CDIP / Tobacco-3482 literature.

See §2.9 for a brief survey of this surface and of mathematical formula extraction (Im2Latex / CROHME / pix2tex) [183] [184] [185] [186] [187].

- **Long-document and multi-page handling** (multi-page reading-order continuity, cross-page table stitching, full-10K-style finance documents). [165] introduces hierarchical multi-page DocVQA; full multi-page parse-benchmark coverage is left to follow-on work.

4.2 Reproducibility caveats

An external team cannot execute the framework end-to-end from this review alone. The principal blockers are as follows:

- **SAVIOR-Bench v1 (n=509)** is an internal artifact and is not redistributable; the D-OCR primary metric `savior_word_recall` is undefined without it.
- **`parse-intent-inhouse-v1` rubric** is internal; the contamination check (Q-DOSS-2) is still open.

- **Qwen3.6-35B-A3B fine-tune endpoint** (private Hyperbots IP) is required for the D-DOWNSTREAM reference extractor.
- **PaIRS commit** (hyprbots/vlm_ocr@1fbbc334) is private; the public reading-order alternatives are reading-order edit distance (DocLayNet) or NED (OmniDocBench).
- **ParseBench n=10 document IDs** are not specified; the registry should publish the exact PDF stems used per split for true reproducibility.

The framework is therefore reproducible *in template* (decoding pins, metric definitions, cell schema) and partially reproducible in *substance* (Apache-2.0 ParseBench, OmniDocBench, FUNSD/CORD/SROIE, the foundational works in Appendix A) on public corpora alone.

4.3 Statistical floor

The first-pass sampled cells use n=10 per split. For a metric on [0,1], this implies an expected 2000-resample bootstrap CI95 half-width of approximately ±0.15, which may obscure engine-to-engine deltas of comparable magnitude (for instance, a published difference of 0.10 PaIRS between two engines is not significant at n=10). No formal power analysis has been performed. The framework's publish-gate (per-document score vectors plus bootstrap CI95) is consistent with the prevailing HELM-era discipline [161]; widening n is the correct next step for headline cells.

4.4 Evaluator-LLM bias note

In D-DOWNSTREAM, the reference extractor (Qwen3.6-35B-A3B) is itself an LLM. This constitutes an LLM-as-judge setting and is therefore susceptible to the judge biases catalogued in the recent LLM-evaluation literature. The framework's countermeasure is the apples-to-apples constraint (a single extractor with identical decoding pins across all upstream parsers), which controls relative ranking but does not eliminate absolute-scale bias.

4.5 Cross-engine fairness note (apples-to-oranges)

The framework's uniform decoding pin `max_tokens=2048` truncates Chandra and Qwen on dense text-content and text-formatting splits in 8 to 10 out of 10 documents. The content-metric cells for those engines under this pin are **labeled apples-to-oranges** and should not be interpreted as engine-capability cells. PaIRS scores on the same documents remain valid, since the metric measures matched-token geometry rather than transcript completeness.

5. Broader Impact

Document parsing sits upstream of two surfaces with social consequences: business-process automation (invoice processing, contract intake, loan underwriting), where parse errors propagate into financial decisions; and digital archives and accessibility (scanning legal, medical, and historical records), where parse errors shape what remains readable to downstream tools. The framework's per-document score vectors and apples-to-apples decoding pins are intended to expose systematic disparities, particularly across languages, document-quality tiers, and document types that are over-represented in upstream training data. Two specific risks are noted. First, evaluation-set contamination: many of the surveyed VLMs were trained on web-scraped corpora that may overlap with the benchmarks on which they are evaluated; Appendix B's contamination column flags known cases but cannot certify the closed-source frontier models. Second, evaluator-LLM bias (§4.4): when a downstream-task evaluator is itself an LLM (here, Qwen3.6-35B-A3B as the frozen extractor for D-DOWNSTREAM), the framework can rank engines reliably under the apples-to-apples constraint but cannot certify absolute capability. Users of the framework are therefore advised to report both the engine under test and the evaluator-LLM pin.

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